

RECOVERY OF MATES ON MARTÍN NORMS AND THE RESONANCE OBSTRUCTION

ABSTRACT. For Martín's renormings p of c_0 (canonical base, finitely or infinitely many blocks) we study the density of norm-attaining operators into ℓ_2^2 through the mate fibres $\mathcal{C}(f) = \{g : (f, g) \text{ is contractive}\}$. Density is equivalent to the statement that every pair $(f, \rho g)$ with $g \in \mathcal{C}(f)$ and $\rho < 1$ is a limit of norm-attaining pairs, and we reduce this to lower limits of the fibres along norm-attaining approximants. We introduce finite certificates with coordinatewise radii and prove that the closed set they generate is transported along every sequence $f_n \rightarrow f$; shifted certificates are transported as well. An averaging theorem over scales, with weights, shows that every mate which is, at every small scale, a certificate of radius comparable to the scale up to a remainder of the same order, belongs to that closed set. This covers weighted directions with unbounded coefficients, all locally split mates, and cross mates carried by off-peak coordinates whose gaps are not summable. At tame supports we identify the sharp second-order invariant, the mass-weighted coefficient Γ_w , and show that it is attained along canonical truncations by means of transfer peaks; as a consequence every tame support without degenerate peaks lies in the recoverable set. On the negative side, we construct an admissible operator T in Martín's construction and a non-attaining first row f at which all intrinsically recoverable mates lie on one line while $\mathcal{C}(f)$ is infinite dimensional (the resonance obstruction), and we show that the canonical norm-attaining truncations do not recover these mates. The density problem itself remains open; we formulate precisely what remains.

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1. SETTING AND STANDING FACTS

All spaces are real. We write $s(t) := \sqrt{1+t^2}$ throughout.

1.1. The base, the blocks and Martín's lemmas. We use the canonical base on $X = c_0$, as in the accompanying note [6]: H is a Hilbert space, $U : H \rightarrow c_0$ is compact with dense range, and

$$B_q = B_{c_0} + U(B_H), \quad q^*(a) = \|a\|_1 + \|U^*a\|_H \quad (a \in \ell_1 = X^*).$$

Since U has dense range, U^* is injective; it is compact, so $\|U^*e_j^*\|_H \rightarrow 0$. The bidual ball is $B_{q^{**}} = B_{\ell_\infty} + U(B_H)$ (the sum of a weak*-compact set and a norm compact set is weak* compact, contains B_q , and is contained in its weak* closure). The norm q is a smoothing of the supremum norm in the sense of Debs, Godefroy and Saint-Raymond [1].

Martín's construction [5] rests on two lemmas, which we recall in the form stated there.

Remark 1.1 (Martín's Lemma A). Let Φ be a sequence of positive numbers with $\|\Phi\|_1 < 1$ and let $D_\Phi : \ell_2 \rightarrow \ell_1$ be the diagonal operator with entries $\Phi(n)$. The norm $|\cdot|_\Phi$ on ℓ_1 with unit ball $B_{\ell_1} + D_\Phi(B_{\ell_2})$ satisfies: (a) $|f|_\Phi^* = \|f\|_\infty + \|D_\Phi f\|_2$ for $f \in \ell_\infty$; (b) $|f|_\Phi^* \geq \|f\|_\infty \geq (1 - \|\Phi\|_1)|f|_\Phi^*$; (c) $|x|_\Phi \leq \|x\|_1 \leq (1 + \|\Phi\|_1)|x|_\Phi$; (d) the dual norm is strictly convex, so $|\cdot|_\Phi$ is smooth; (e) if $|f|_\Phi^* = 1$, $\langle f, x \rangle = |x|_\Phi$ and $|x(n)| > \Phi(n)|x|_\Phi$, then $f(n) = \text{sign}(x(n))\|f\|_\infty$.

Remark 1.2 (Martín's Lemma B and the choice of T). Lemma B of [5] reads: *Let Y be a separable operator range. Then there is a norm-one injective operator $T : \ell_1(\mathbb{N} \times \mathbb{N}) \rightarrow Y$ such that for every m the set $\{T(e_{n,m})/\|T(e_{n,m})\| : n \in \mathbb{N}\}$ is dense in S_Y .* Its proof takes a bijection $\sigma : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, a sequence (w'_n) in S_Y such that every point of S_Y is an accumulation point of (w'_n) , and repeats the proof of [4, Proposition 2.8] with the sequence $w_{\sigma(n,m)} = w'_n$. Two features matter below. First, the sequence (w'_n) is *arbitrary* subject to the accumulation property, so “Martín's space” is a family of spaces indexed by the admissible choices of T . Second, every block m follows the same sequence (w'_n) , so that near duplicates $Te_{n,m}/\|Te_{n,m}\| \approx Te_{n,m'}/\|Te_{n,m'}\|$ across blocks are built into the construction

(up to the perturbations produced in [4]). No other property of T is used in [5], and no other property will be used here unless it is stated explicitly.

Definition 1.3 (Admissible operators). An operator $T : \ell_1(\mathbb{N} \times \mathbb{N}) \rightarrow X^* = \ell_1$ is *admissible* if

- (T-a) $q^*(Te_{n,m}) \leq 1$ for all (n, m) , with equality for some (n, m) ;
- (T-b) T is injective;
- (T-c) $Y := T(\ell_1(\mathbb{N} \times \mathbb{N}))$ satisfies $Y \cap c_{00} = \{0\}$;
- (T-d) for every m , the set $\{u_{n,m} : n \in \mathbb{N}\}$, where $u_{n,m} := Te_{n,m}/q^*(Te_{n,m})$, is norm dense in S_{q^*} .

Since $\text{NA}((X, q), \mathbb{R}) = c_{00}$ (Proposition 1.9(c) below), (T-c) is the condition $Y \cap \text{NA}(q) = \{0\}$ of [5]. As S_{q^*} has no isolated points, (T-d) implies that every tail $\{u_{n,m} : n \geq n_0\}$ is dense in S_{q^*} .

1.2. The norm p . Fix an admissible T . For $m \in \mathbb{N}$ put

$$\begin{aligned} \Phi_m(k) &:= 2^{-m-k} q^*(Te_{k,m}) \leq 2^{-m-k}, & \lambda_{k,m} &:= m\Phi_m(k), \\ (R_mx)(k) &:= \lambda_{k,m} u_{k,m}(x) & (x \in X). \end{aligned}$$

Let D_m be the diagonal operator with entries $\Phi_m(k)$, let $|\cdot|_m$ be the norm of Remark 1.1 for $\Phi = \Phi_m$ on $V_m = \ell_1$, and $N_m(w) := |w|_m^* = \|w\|_\infty + \|D_m w\|_2$. For a nonempty set $I \subseteq \mathbb{N}$ of *blocks* put

$$\begin{aligned} V &:= \left(\bigoplus_{m \in I} V_m \right)_{\ell_1}, & V^* &= \left(\bigoplus_{m \in I} (\ell_\infty, N_m) \right)_{\ell_\infty}, \\ Lx &:= (R_mx)_{m \in I}, & p(x) &:= q(x) + \|Lx\|_V. \end{aligned}$$

For $I = \mathbb{N}$ this is Martín's norm (with the canonical base); for $I = \{1, \dots, N\}$ we obtain the finite-block norms p_N of [7]. Since $|(R_mx)(k)| \leq m2^{-m-k}q(x)$, $\|R_m\| \leq m2^{-m}$; each R_m is a norm limit of finite-rank operators, hence compact, and L is compact. For $w \in V^*$,

$$(1) \quad L^*w = \sum_m R_m^* w_m = T \left(\sum_{m \in I} \sum_k m2^{-m-k} w_m(k) e_{k,m} \right) \in Y,$$

an absolutely convergent series; in particular L^* is injective on V^* . For $\theta \in X^{**} = \ell_\infty$, $(R_m^{**}\theta)(k) = \lambda_{k,m}\theta(u_{k,m})$, so $R_m^{**}(X^{**}) \subseteq \ell_1$.

Remark 1.4 (Finite and infinite block sets). By the remark on Martín tails in [7], if the set $\text{NA}((c_0, p_N), F)$ is dense in $\mathcal{L}((c_0, p_N), F)$ for arbitrarily large N , then $\text{NA}((c_0, p), F)$ is dense for Martín's p ($I = \mathbb{N}$): one has $p = p_N + s_N$ with $s_N \leq \eta_N q \leq \eta_N p_N$, $\eta_N \rightarrow 0$, and the lift lemma of [7] applies. Hence finite block sets suffice for the density problem. Sections 1–4 are valid for every nonempty I (certificates involve finitely many blocks); the one place where $I = \mathbb{N}$ needs an extra hypothesis is Corollary 4.3. Sections 5 and 6 assume that I is finite.

1.3. Elementary structure.

Lemma 1.5 (Dual ball). $B_{p^*} = B_{q^*} + L^*(B_{V^*})$, and for every $h \in X^*$

$$p^*(h) = \min\{\max(q^*(A), \|W\|_{V^*}) : A + L^*W = h\},$$

the minimum being attained. Moreover $p^{**}(\theta) = q^{**}(\theta) + \|L^{**}\theta\|_V$ for $\theta \in X^{**}$, and $L^{**}(X^{**}) \subseteq V$.

Proof. If $q^*(A) \leq 1$ and $\|W\| \leq 1$ then $(A + L^*W)(x) \leq q(x) + \|Lx\| = p(x)$. The set $B_{q^*} + L^*(B_{V^*})$ is convex and weak* compact (L^* is weak*-continuous), and its support function on X is $q(x) + \|Lx\|_V = p(x)$, which is also the support function of B_{p^*} ; two weak*-closed convex sets with the same support function on X coincide. The formula for p^* follows by homogeneity and compactness. Taking suprema over B_{p^*} gives the bidual formula; L^{**} maps into V because L is compact. $\square$

Lemma 1.6 (Block threshold [7, Lemma 5.6]). *Fix a block and drop the index m . Let $\zeta \in \ell_1 \setminus \{0\}$ and $w = J(\zeta)$ its norming functional ($N(w) = 1$, $w(\zeta) = |\zeta|$). Put $M = \|w\|_\infty$, $C = \|Dw\|_2$ and $P = \{k : |w(k)| = M\}$. Then $M > 0$, $C > 0$, $M + C = 1$, and*

$$\zeta/|\zeta| = \alpha + D^2w/C, \quad \|\alpha\|_1 = 1, \quad \text{supp } \alpha \subseteq P, \quad \text{sign } \alpha(k) = \text{sign } w(k) \quad (k \in \text{supp } \alpha).$$

In particular $P \neq \emptyset$; off P one has $w(k) = C\zeta(k)/(\Phi(k)^2|\zeta|)$, and $|\zeta(k)| > |\zeta|\Phi(k)^2M/C$ implies $k \in P$ and $w(k) = M \text{sign } \zeta(k)$.

Proof. $w \neq 0$ and D is injective, so $M, C > 0$. The ball $B_{\ell_1} + D(B_{\ell_2})$ is closed (D is compact), so $\zeta/|\zeta| = \alpha + D\beta$ with $\|\alpha\|_1, \|\beta\|_2 \leq 1$. Then $1 = \langle w, \alpha \rangle + \langle Dw, \beta \rangle \leq M\|\alpha\|_1 + C\|\beta\|_2 \leq M + C = 1$, so $\langle w, \alpha \rangle = M = M\|\alpha\|_1$ and $\beta = Dw/C$; the first equality forces $\|\alpha\|_1 = 1$ and α to live on P with the signs of w . The remaining claims follow by reading the identity coordinatewise. $\square$

For a block vector $\zeta = R_m^{**}\theta$ the condition of the last sentence reads $|\theta(u_{k,m})| > \vartheta_m \Phi_m(k)$ with the *threshold constant* $\vartheta_m := |\zeta|_m M_m / (m C_m)$. For $k \in P$ we define the *margin* $\mu_{k,m} := |\theta(u_{k,m})| - \vartheta_m \Phi_m(k) \geq 0$; reading Lemma 1.6 at $k \in P$ gives the identity

$$(2) \quad |\zeta|_m |\alpha_m(k)| = \lambda_{k,m} \mu_{k,m} \quad (k \in P_m).$$

Peaks with $\alpha_m(k) = 0$ (equivalently $\mu_{k,m} = 0$) are called *degenerate*. We write $Q_m := \mathbb{N} \setminus P_m$ (strict non-peaks) and $\text{gap}_m(k) := M_m - |w_m(k)|$.

Lemma 1.7 (Rigidity and independence). (a) *If $b_+ + L^*W_+ = b_- + L^*W_-$ with $b_+ - b_- \in c_{00}$ and $W_\pm \in V^*$, then $b_+ = b_-$ and $W_+ = W_-$.* (b) *Let $J \subseteq \mathbb{N}$ be cofinite, I finite, and for each $m \in I$ let $\Lambda_m \subseteq \mathbb{N}$ be finite and $P'_m \subseteq \mathbb{N}$ be infinite, $s \in \{\pm 1\}^{P'_m}$. Put $\pi_m := \sum_{k \in P'_m} s_k \lambda_{k,m} u_{k,m}$. Then the restrictions to $c_{00}(J)$ of the functionals $u_{k,m}$ ($k \in \Lambda_m$, $m \in I$) and π_m ($m \in I$) are linearly independent; consequently $\nu \mapsto (u_{k,m}(\nu), \pi_m(\nu))$ maps $c_{00}(J)$ onto the corresponding finite-dimensional space.*

Proof. (a) $b_+ - b_- = L^*(W_- - W_+) \in Y \cap c_{00} = \{0\}$, and L^* is injective. (b) A vanishing combination $\sum c_{k,m} u_{k,m} + \sum_m d_m \pi_m$ on $c_{00}(J)$ is an element of Y supported in the finite set $\mathbb{N} \setminus J$, hence zero by (T-c). By (T-b) its coefficient at every $e_{k,m}$ vanishes: at the infinitely many $k \in P'_m \setminus \Lambda_m$ this coefficient is $d_m s_k m 2^{-m-k}$, so $d_m = 0$, and then $c_{k,m} = 0$. $\square$

Proposition 1.8 (Strict convexity and forced decomposition). (a) *Each R_m^{**} is injective, and p^{**} is strictly convex.*

(b) *Every $f \in S_{p^*}$ has a unique normer $\xi \in S_{p^{**}}$. Put $q_0 := q^{**}(\xi)$. Then $0 < q_0 < 1$, $\|L^{**}\xi\|_V = 1 - q_0$, every $R_m^{**}\xi \neq 0$, and there is exactly one decomposition $f = a + L^*w$ with $q^*(a) \leq 1$ and $\|w\|_{V^*} \leq 1$; it satisfies*

$$q^*(a) = 1, \quad a(\xi) = q_0, \quad w_m = J_m(R_m^{**}\xi), \quad N_m(w_m) = 1 \quad (m \in I).$$

(c) *Put $\nu := \|U^*a\|_H > 0$ and $e := U^*a/\nu$. Then $\hat{z} := \xi/q_0 = z + Ue$ with $z \in B_{\ell_\infty}$ and $z_j = \text{sign } a_j$ on $F := \text{supp } a$; and $a(\hat{z}) = 1$, $b(\hat{z}) = b(z) + \langle U^*b, e \rangle$ for $b \in \ell_1$.*

Proof. (a) If $\theta(u_{k,m}) = 0$ for all k , then θ vanishes on a dense subset of S_{q^*} , so $\theta = 0$. Let $p^{**}(\xi) = p^{**}(\eta) = p^{**}((\xi+\eta)/2) = 1$. Since q^{**} and each $|R_m^{**} \cdot|_m$ are seminorms whose sum is p^{**} (Lemma 1.5), we get $|R_m^{**}(\xi+\eta)|_m = |R_m^{**}\xi|_m + |R_m^{**}\eta|_m$ for every m . Fix m ; $\zeta_1 = R_m^{**}\xi$ and $\zeta_2 = R_m^{**}\eta$ are nonzero, and a norming functional w of $\zeta_1 + \zeta_2$ norms both. If ξ, η were not positively proportional there would be $v \in S_{q^*}$ with $\xi(v) > 0 > \eta(v)$ (use surjectivity of $v \mapsto (\xi(v), \eta(v))$ if they are independent). Choose $n_j \rightarrow \infty$ with $u_{n_j,m} \rightarrow v$. Since $\Phi_m(n_j) \rightarrow 0$, Lemma 1.6 applied to ζ_1 and to ζ_2 gives $w(n_j) = \|w\|_\infty$ and $w(n_j) = -\|w\|_\infty$ for large j , which is absurd. Hence $\eta = \lambda\xi$ with $\lambda > 0$, and $\lambda = 1$.

(b) The normers of f form a convex subset of $S_{p^{**}}$, hence a point. For any decomposition $f = a + L^*w$ with $q^*(a), \|w\| \leq 1$, $1 = f(\xi) = a(\xi) + w(L^{**}\xi) \leq q^{**}(\xi) + \|L^{**}\xi\| = 1$, so $a(\xi) = q_0$ and $w_m(R_m^{**}\xi) = |R_m^{**}\xi|_m$ for each m . As $R_m^{**}\xi \neq 0$ and $|\cdot|_m$ is smooth, $w_m = J_m(R_m^{**}\xi)$ is determined, hence so is $a = f - L^*w$. Finally $q_0 > 0$ because q^{**} is a norm and $\xi \neq 0$, $q_0 < 1$ because $L^{**}\xi \neq 0$, and $q_0 = a(\xi) \leq q^*(a)q_0$ gives $q^*(a) = 1$.

(c) $\xi/q_0 \in B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, so $\xi/q_0 = z + Uh$ with $\|z\|_\infty, \|h\| \leq 1$; then $1 = a(z) + \langle U^*a, h \rangle \leq \|a\|_1 + \|U^*a\| = 1$ forces $a(z) = \|a\|_1$ and $h = e$. $\square$

Throughout, the *forced data* of $f \in S_{p^*}$ are $(\xi, q_0, a, w, F, z, \hat{z}, e, \nu)$ and, for each block, $\zeta_m := R_m^{**}\xi$, $\sigma_m := |\zeta_m|_m$, $M_m, C_m, P_m, \alpha_m, Q_m, \text{gap}_m, \vartheta_m, \mu_{k,m}$. Note $q_0 + \sum_m \sigma_m = 1$. The set $K := \{j \notin F : |z_j| = 1\}$ is the set of *contacts* (kinks) and $J := \mathbb{N} \setminus (F \cup K)$ the set of *free coordinates*; the *room* of $j \in J$ is $1 - |z_j| > 0$.

1.4. Smoothness and norm-attaining functionals.

Proposition 1.9 (Fréchet smoothness). (a) *The norm p is Fréchet differentiable at every $x \neq 0$, and $\nabla p(x) = \nabla q(x) + L^*J_V(Lx)$, where $J_V(Lx) = (J_m(R_mx))_m$.*

(b) *$\text{NA}((X, p), \mathbb{R}) \cap S_{p^*} = \nabla p(S_p)$, and the duality map $\nabla p : S_p \rightarrow S_{p^*}$ is norm-to-norm continuous.*

(c) *$\text{NA}((X, q), \mathbb{R}) = c_{00}$. More precisely, if $a' \in c_{00}$, $q^*(a') = 1$, and $z' \in c_0$ satisfies $\|z'\|_\infty \leq 1$ and $z'_j = \text{sign } a'_j$ on $\text{supp } a'$, then $y' := z' + U(U^*a'/\|U^*a'\|)$ satisfies $q(y') = 1$, $\nabla q(y') = a'$ and $\nabla p(y') = a' + L^*J_V(Ly')$. Conversely every $f' \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ equals $\nabla p(y')$ for such a pair (a', z') , with $a' = \nabla q(y')$.*

Proof. *Gateaux smoothness.* The norm q^* is strictly convex: if $q^*(a) = q^*(b) = 1 = q^*(\frac{a+b}{2})$, then $\|U^*a + U^*b\| = \|U^*a\| + \|U^*b\|$, so $U^*b = \lambda U^*a$ with $\lambda \geq 0$ (both are nonzero), hence $b = \lambda a$ by injectivity of U^* , and $\lambda = 1$. Thus q is Gateaux smooth. In the same way N_m is strictly convex (D_m is injective), so $|\cdot|_m$ is smooth. Let $x \neq 0$ and let $f, f' \in S_{p^*}$ both norm x . Write $f = A + L^*W$ with $q^*(A), \|W\| \leq 1$ (Lemma 1.5). Then $p(x) = A(x) + W(Lx) \leq q(x) + \|Lx\| = p(x)$ forces $A(x) = q(x)$ and $W_m(R_mx) = |R_mx|_m$ for every m . Since R_m is injective (by the argument of Proposition 1.8(a)), $R_mx \neq 0$, so $A = \nabla q(x)$ and $W_m = J_m(R_mx)$. The same holds for f' , so $f = f'$.

Fréchet smoothness. By Šmul'yan's lemma it suffices to show: if $x \in S_p$, $f_n \in B_{p^*}$ and $f_n(x) \rightarrow 1$, then $f_n \rightarrow \nabla p(x)$ in norm. Write $f_n = a_n + L^*w_n$ with $q^*(a_n), \|w_n\| \leq 1$. As $a_n(x) \leq q(x)$, $w_n(Lx) \leq \|Lx\|$ and the sum tends to $p(x)$, we get $a_n(x) \rightarrow q(x)$. Write $x/q(x) = z + Uh$ with $z \in B_{c_0}$, $\|h\| \leq 1$. Then

$$\frac{a_n(x)}{q(x)} \leq \sum_j |a_n(j)| |z_j| + \|U^*a_n\| \leq 1 - \sum_j |a_n(j)| (1 - |z_j|),$$

so $\sum_j |a_n(j)|(1 - |z_j|) \rightarrow 0$. The set $G = \{j : |z_j| \geq \frac{1}{2}\}$ is finite, $\sum_{j \notin G} |a_n(j)| \rightarrow 0$, and the coordinates in G are bounded; hence every subsequence of (a_n) has a norm convergent subsequence, whose limit a' satisfies $q^*(a') \leq 1$, $a'(x) = q(x)$, so $a' = \nabla q(x)$. Every subsequence of (w_n) has a weak* convergent subsequence $w_n \rightarrow w$, and $L^*w_n \rightarrow L^*w$ in norm because L is compact. Then $\nabla q(x) + L^*w \in B_{p^*}$ norms x , so it equals $\nabla p(x)$. This proves (a), and the continuity in (b).

(b) If $f \in S_{p^*}$ attains its norm at $x \in S_p$, then $f = \nabla p(x)$.

(c) If $a \in S_{q^*}$ attains its norm at $y \in S_q$, write $y = z + Uh$, $z \in B_{c_0}$, $\|h\| \leq 1$; then $1 = a(z) + \langle U^*a, h \rangle$ forces $|z_j| = 1$ on $\text{supp } a$, which is finite because $z \in c_0$. For the pair (a', z') : $y' \in B_q$ and $a'(y') = \|a'\|_1 + \|U^*a'\| = 1 = q^*(a')$, hence $q(y') = 1$ and $\nabla q(y') = a'$; then (a) gives $\nabla p(y')$. Conversely, if $f' = \nabla p(x')$ with $x' \in S_p$, then $a' := \nabla q(x') \in c_{00}$ and $y' := x'/q(x') = z' + Uh'$ with $z' \in B_{c_0}$; the equality $1 = a'(z') + \langle U^*a', h' \rangle \leq \|a'\|_1 + \|U^*a'\| = 1$ forces $z' = \text{sign } a'$ on $\text{supp } a'$ and $h' = U^*a'/\|U^*a'\|$; finally $\nabla p(y') = \nabla p(x')$. $\square$

Proposition 1.10 (Continuity of the forced data). *Let $f_n \rightarrow f$ in S_{p^*} (not necessarily norm attaining) with forced data indexed by n . Then $\xi_n \rightarrow \xi$ weak*; $R_m^{**}\xi_n \rightarrow R_m^{**}\xi$ in ℓ_1 ; $w_{n,m} \rightarrow w_m$ weak* and $D_m w_{n,m} \rightarrow D_m w_m$ in ℓ_2 , so $M_{n,m} \rightarrow M_m$, $C_{n,m} \rightarrow C_m$, $\sigma_{n,m} \rightarrow \sigma_m$ and $\vartheta_{n,m} \rightarrow \vartheta_m$; $R_m^* w_{n,m} \rightarrow R_m^* w_m$ and $L^* w_n \rightarrow L^* w$ in norm; $a_n \rightarrow a$ in ℓ_1 ; $e_n \rightarrow e$ in H ; $q_0^{(n)} \rightarrow q_0$; $\hat{z}_n \rightarrow \hat{z}$ and $z_n \rightarrow z$ weak*.*

Proof. A weak* cluster point η of (ξ_n) lies in $B_{p^{**}}$ and $f(\eta) = \lim f_n(\xi_n) = 1$, so $\eta = \xi$; $B_{p^{**}}$ is weak* metrizable. R_m is compact, so R_m^{**} maps bounded weak* convergent sequences to norm convergent ones. A weak* cluster point w' of $(w_{n,m})_n$ has $N_m(w') \leq 1$ and $w'(\zeta_m) = \lim w_{n,m}(\zeta_{n,m}) = \sigma_m$, so $w' = w_m$. Since $\Phi_m \in \ell_2$, D_m maps bounded weak* convergent sequences to norm convergent ones; $M = 1 - C$. L^* is weak*-to-norm continuous on bounded sets, and $a_n = f_n - L^*w_n$. Finally $q_0^{(n)} = a_n(\xi_n) \rightarrow a(\xi)$, $e_n = U^*a_n/\|U^*a_n\|$, and $z_n = \xi_n/q_0^{(n)} - Ue_n$. $\square$

Proposition 1.11 (Norm-attaining approximants). *Let $f \in S_{p^*}$ have forced data as above. For a sequence $f'_n \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ the following are equivalent:*

- (i) $f'_n \rightarrow f$ in norm;
- (ii) $f'_n = \nabla p(y'_n)$ with $y'_n = z'_n + U(U^*a'_n/\|U^*a'_n\|)$, where $a'_n \in c_{00} \cap S_{q^*}$, $a'_n \rightarrow a$ in ℓ_1 , $z'_n \in c_0$, $\|z'_n\|_\infty \leq 1$, $z'_n = \text{sign } a'_n$ on $\text{supp } a'_n$, and $z'_n \rightarrow z$ coordinatewise.

In particular, for every n the coordinates of z'_n outside a finite window may be chosen freely (moduli ≤ 1 , membership in c_0) as long as the windows exhaust $\mathbb{N}$.

Proof. (ii) $\Rightarrow$ (i): $y'_n \rightarrow \hat{z}$ weak* (bounded coordinatewise convergence plus norm convergence of the U -part, as $U^*a'_n/\|U^*a'_n\| \rightarrow e$). By compactness of L , $p(y'_n) = 1 + \|Ly'_n\| \rightarrow 1 + \|L^*\hat{z}\| = 1/q_0$, so $x_n := y'_n/p(y'_n) \rightarrow \xi$ weak*. Then $\nabla q(x_n) = a'_n \rightarrow a$, and $L^*J_V(Lx_n) \rightarrow L^*w$ in norm by the argument of Proposition 1.10 (a weak* cluster point of $J_V(Lx_n)$ norms $L^{**}\xi$, whose components are nonzero). (i) $\Rightarrow$ (ii): write $f'_n = \nabla p(x'_n)$ (Proposition 1.9); by Proposition 1.10, $x'_n \rightarrow \xi$ weak*, $a'_n := \nabla q(x'_n) \rightarrow a$, and Proposition 1.9(c) gives the form of $y'_n = x'_n/q(x'_n)$; finally $z'_n = y'_n - U(\cdot) \rightarrow z$ weak*. $\square$

1.5. Mates and the reduction. For $f \in S_{p^*}$ we use, as in [6],

$$\mathcal{C}(f) := \{g \in X^* : p^*(f + tg) \leq s(t) \text{ for all } t \in \mathbb{R}\}.$$

Proposition 1.12 (Reduction). (a) For $f \in S_{p^*}$ and $g \in X^*$ the operator $(f, g) : (X, p) \rightarrow \ell_2^2$ is contractive iff $g \in \mathcal{C}(f)$ iff $f(y)^2 + g(y)^2 \leq p(y)^2$ for all $y \in X$. (b) Every $S \in \mathcal{L}((X, p), \ell_2^2)$ of norm one is, after a rotation of ℓ_2^2 , of the form (f, g) with $f \in S_{p^*}$ and $g \in \mathcal{C}(f)$. Hence $\text{NA}((X, p), \ell_2^2)$ is dense iff $(f, \rho g) \in \text{cl NA}((X, p), \ell_2^2)$ for all $f \in S_{p^*}$, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. (c) If $g \in \mathcal{C}(f)$ then $p^*(g) \leq 1$, $g(\xi) = 0$, and $p^*(f + t\rho g) \leq s(\rho t) < s(t)$ for $t \neq 0$, $\rho \in (0, 1)$. (d) If f attains its norm at x and $g \in \mathcal{C}(f)$, then (f, g) attains its norm at x .

Proof. (a) $\|(f, g)\| = \sup_{\theta} p^*(\cos \theta f + \sin \theta g)$ and $p^*(\cos \theta f + \sin \theta g) = |\cos \theta| p^*(f + \tan \theta g)$ with $|\cos \theta| = s(\tan \theta)^{-1}$; the case $\cos \theta = 0$ is the limit $t \rightarrow \infty$. The primal form is $\|(f, g)\|^2 = \sup_{p(y) \leq 1} f(y)^2 + g(y)^2$. (b) $\theta \mapsto p^*(S^*(\cos \theta, \sin \theta))$ attains its maximum 1; rotate that direction to e_1 . Since NA is a cone and $(f, g) = \lim_{\rho \rightarrow 1} (f, \rho g)$, the reduction follows. (c) Divide $p^*(f + tg) \leq s(t)$ by $|t| \rightarrow \infty$; and $1 + tg(\xi) = (f + tg)(\xi) \leq s(t)$ for all t forces $g(\xi) = 0$. (d) $g(x)^2 \leq p(x)^2 - f(x)^2 = 0$ (this observation is from [3]). $\square$

Proposition 1.13 (Primal description of the fibre). For $f \in S_{p^*}$ put $r_f := \sqrt{p^2 - f^2}$ on X and let $\tilde{r}_f$ be its largest sublinear minorant, $\tilde{r}_f(x) = \inf\{\sum_j r_f(y_j) : \sum_j y_j = x\}$ (finite families). Then $\tilde{r}_f$ is a seminorm, $\mathcal{C}(f) = \{g \in X^* : g \leq \tilde{r}_f\}$, $\mathcal{C}(f)$ is convex, symmetric and weak* compact, and $\max_{g \in \mathcal{C}(f)} g(x) = \tilde{r}_f(x)$ for $x \in X$.

Proof. The infimum φ is subadditive, positively homogeneous, even and $\leq r_f$, and every sublinear minorant s of r_f satisfies $s(x) \leq \sum s(y_j) \leq \sum r_f(y_j)$, so $\varphi = \tilde{r}_f$. A linear g satisfies $g \leq r_f$ iff $g \leq \tilde{r}_f$, and as r_f is even this is $|g| \leq r_f$, i.e. $g \in \mathcal{C}(f)$ by Proposition 1.12(a). Given x , the Hahn–Banach theorem gives a linear $g \leq \tilde{r}_f$ with $g(x) = \tilde{r}_f(x)$, continuous because $\tilde{r}_f \leq p$. $\square$

Definition 1.14 (Recovery). For subsets $C_n \subseteq X^*$ let $\text{Li}_n C_n := \{g : \text{dist}(g, C_n) \rightarrow 0\}$ (Kuratowski lower limit; it is closed, and convex if the C_n are convex). A mate $g \in \mathcal{C}(f)$ is *recovered along* a sequence $f_n \rightarrow f$ in S_{p^*} if $g \in \text{Li}_n \mathcal{C}(f_n)$. We write

$$\mathcal{R} := \{f \in S_{p^*} : (f, g) \in \text{cl NA}((X, p), \ell_2^2) \text{ for every } g \in \mathcal{C}(f)\}$$

for the *recoverable set*. By Proposition 1.12, $\text{NA}((X, p), \ell_2^2)$ is dense iff $\mathcal{R} = S_{p^*}$.

Lemma 1.15 (Recoverability of a single pair). Let $f \in S_{p^*}$, $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. Then $(f, \rho g) \in \text{cl NA}((X, p), \ell_2^2)$ iff there are $f_n \in \text{NA}((X, p), \mathbb{R}) \cap S_{p^*}$ with $f_n \rightarrow f$ and $\rho g \in \text{Li}_n \mathcal{C}(f_n)$.

Proof. If $g_n \in \mathcal{C}(f_n)$ and $g_n \rightarrow \rho g$, then (f_n, g_n) attains its norm (Proposition 1.12(d)) and converges to $(f, \rho g)$. Conversely let $S_k \in \text{NA}$ converge to $T := (f, \rho g)$; $\|T\| = 1$. For a unit vector $y = (\cos \theta, \sin \theta)$ with $\sin \theta \neq 0$ we have $p^*(T^*y) < 1$: if $\cos \theta \neq 0$, $p^*(T^*y) = |\cos \theta| p^*(f + \tan \theta \rho g) \leq |\cos \theta| s(\rho \tan \theta) < 1$, and $p^*(\rho g) \leq \rho < 1$ otherwise. Let S_k attain its norm at $x_k \in S_p$ and $y_k := S_k x_k / \|S_k x_k\|$. Then $p^*(T^*y_k) \geq \|S_k\| - \|S_k - T\| \rightarrow 1$, so every limit point of (y_k) is $\pm e_1$; replacing x_k by $-x_k$ we may assume $y_k \rightarrow e_1$. Let Q_k be the rotation with $Q_k y_k = e_1$; then $Q_k \rightarrow I$, and $Q_k S_k = (\tilde{f}_k, \tilde{g}_k)$ attains its norm at x_k with $\tilde{f}_k(x_k) = \|S_k\|$, $\tilde{g}_k(x_k) = 0$. The functionals $f_k := \tilde{f}_k / \|S_k\|$ and $g_k := \tilde{g}_k / \|S_k\|$ satisfy $\|(f_k, g_k)\| = 1 = f_k(x_k)$, so $f_k \in \text{NA} \cap S_{p^*}$, $g_k \in \mathcal{C}(f_k)$, and $(f_k, g_k) \rightarrow (f, \rho g)$. $\square$

2. THE LOWER-LIMIT THEOREM

Theorem 2.1 (Global compactness [6, Theorem 2.1]). *If $f_n \rightarrow f$ in S_{p^*} and $g_n \in \mathcal{C}(f_n)$, then (g_n) has a norm convergent subsequence and every limit of such a subsequence lies in $\mathcal{C}(f)$. Consequently every $\mathcal{C}(f)$ is norm compact, $f \mapsto \mathcal{C}(f)$ is upper semicontinuous, and $\mathcal{C}(f) \cup \bigcup_n \mathcal{C}(f_n)$ is norm compact.*

Proof. $p^*(g_n) \leq 1$; pass to a weak* convergent subsequence $g_n \rightarrow g$. Weak* lower semicontinuity of p^* gives $g \in \mathcal{C}(f)$. Let $\Delta := \limsup_n \|g_n - g\|_1$, realized along a further subsequence. Fix $t > 0$ and write $f_n + tg_n = A_n + L^*W_n$ with $q^*(A_n), \|W_n\| \leq s(t)$ (Lemma 1.5). Passing to a subsequence, $L^*W_n \rightarrow k$ in norm, $W_n \rightarrow W$ weak*, and $A_n \rightarrow A := f + tg - k$ weak*. Since $k(\xi) = W(L^{**}\xi) \leq s(t)(1 - q_0)$ we get $A(\xi) \geq 1 - s(t)(1 - q_0)$ and $q^*(A) \geq A(\xi)/q_0$. Moreover $U^*A_n \rightarrow U^*A$ in norm, and for bounded coordinatewise convergent sequences in ℓ_1 one has $\|A_n\|_1 - \|A_n - A\|_1 \rightarrow \|A\|_1$. Since $t(g_n - g) = (A_n - A) + (L^*W_n - k) - (f_n - f)$,

$$t\Delta = \limsup_n \|A_n - A\|_1 = \limsup_n (q^*(A_n) - q^*(A)) \leq s(t) - \frac{1 - s(t)(1 - q_0)}{q_0} = \frac{s(t) - 1}{q_0}.$$

Letting $t \rightarrow 0$ gives $\Delta = 0$. The remaining assertions follow (for the union, a sequence either has infinitely many terms in one compact fibre or a subsequence to which the first part applies). $\square$

Theorem 2.2 (Lower-limit theorem). *Let $f_n \rightarrow f$ in S_{p^*} (not necessarily norm attaining) and $\rho \in [0, 1]$. The following are equivalent:*

- (i) $\rho\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$;
- (ii) $\liminf_n \tilde{r}_{f_n}(x) \geq \rho \tilde{r}_f(x)$ for every $x \in X$;
- (iii) (ii) holds for all x in a dense subset of X .

Proof. (i) $\Rightarrow$ (ii): choose $g \in \mathcal{C}(f)$ with $g(x) = \tilde{r}_f(x)$ (Proposition 1.13) and $g_n \in \mathcal{C}(f_n)$ with $g_n \rightarrow \rho g$; then $\tilde{r}_{f_n}(x) \geq g_n(x) \rightarrow \rho \tilde{r}_f(x)$. (ii) $\Leftrightarrow$ (iii): each $\tilde{r}_h$ is a seminorm bounded by p , so $|\tilde{r}_h(x) - \tilde{r}_h(y)| \leq p(x - y)$ uniformly in h . (ii) $\Rightarrow$ (i): suppose $g \in \rho\mathcal{C}(f)$ and $\text{dist}(g, \mathcal{C}(f_{n_j})) \geq \delta > 0$. All $\mathcal{C}(f_{n_j})$ lie in the compact set of Theorem 2.1; by Blaschke's selection theorem we may assume $\mathcal{C}(f_{n_j}) \rightarrow A$ in the Hausdorff metric. Then A is compact and convex and $\text{dist}(g, A) \geq \delta$. For $x \in X$ the support functions satisfy $|h_C(x) - h_{C'}(x)| \leq d_H(C, C')\|x\|_\infty$, so $h_A(x) = \lim_j \tilde{r}_{f_{n_j}}(x) \geq \rho \tilde{r}_f(x) \geq g(x)$. As A is weak* compact and convex, $g \notin A$ would give $x \in c_0$ with $g(x) > h_A(x)$. Contradiction. $\square$

Corollary 2.3 (Finite, diagonal and mate versions). *For $f \in S_{p^*}$ the following are equivalent:*

- (a) there are $f_n \in \text{NA} \cap S_{p^*}$ with $f_n \rightarrow f$ and $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$;
- (b) for every $\varepsilon > 0$, $\rho < 1$ and $x_1, \dots, x_k \in X$ there is $f' \in \text{NA} \cap S_{p^*}$ with $p^*(f' - f) < \varepsilon$ and $\tilde{r}_{f'}(x_i) \geq \rho \tilde{r}_f(x_i) - \varepsilon$ for $i \leq k$;
- (c) for every $\varepsilon > 0$, $\rho < 1$ and $g_1, \dots, g_k \in \mathcal{C}(f)$ there is $f' \in \text{NA} \cap S_{p^*}$ with $p^*(f' - f) < \varepsilon$ and $\text{dist}(\rho g_i, \mathcal{C}(f')) < \varepsilon$ for $i \leq k$;
- (d) as (c), uniformly: $\sup_{g \in \mathcal{C}(f)} \text{dist}(\rho g, \mathcal{C}(f')) < \varepsilon$.

If they hold, then $f \in \mathcal{R}$.

Proof. (a) $\Rightarrow$ (d): $\text{dist}(g, \mathcal{C}(f_n)) \rightarrow 0$ for each g , uniformly on the compact set $\mathcal{C}(f)$ (finite nets and 1-Lipschitz dependence on g); and $\text{dist}(\rho g, \mathcal{C}(f_n)) \leq \rho \text{dist}(g, \mathcal{C}(f_n))$ because $\mathcal{C}(f_n)$ is convex and contains 0. (d) $\Rightarrow$ (c) is trivial. (c) $\Rightarrow$ (b): pick $g_i \in \mathcal{C}(f)$

with $g_i(x_i) = \tilde{r}_f(x_i)$ and apply (c) with $\varepsilon/\max(1, \max_i \|x_i\|)$. (b) $\Rightarrow$ (a): let (x_i) be dense in X ; for each n apply (b) with $\varepsilon = 1/n$, $\rho = 1 - 1/n$ and $x_1, \dots, x_n$ to obtain f_n . Then $\liminf_n \tilde{r}_{f_n}(x_i) \geq \tilde{r}_f(x_i)$ for every i , and Theorem 2.2 with $\rho = 1$ gives (a). The last claim is Lemma 1.15. $\square$

Theorem 2.4 (Transitivity). $\text{NA} \cap S_{p^*} \subseteq \mathcal{R}$. Let $f \in S_{p^*}$ and suppose that for every $\rho < 1$ there is a sequence $f_n^\rho \in \mathcal{R}$ with $f_n^\rho \rightarrow f$ and $\liminf_n \tilde{r}_{f_n^\rho}(x) \geq \rho \tilde{r}_f(x)$ on a dense set of $x \in X$. Then $f \in \mathcal{R}$.

Proof. The first claim is Proposition 1.12(d). By Theorem 2.2, $\rho\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n^\rho)$. Given $g \in \mathcal{C}(f)$ choose $g_n \in \mathcal{C}(f_n^\rho)$ with $g_n \rightarrow \rho g$; each (f_n^ρ, g_n) lies in clNA , hence so does $(f, \rho g)$; let $\rho \rightarrow 1$. $\square$

Remark 2.5 (Several rows and the residual set). For $d \geq 1$ let $\mathcal{C}_d(f) := \{(g_1, \dots, g_d) : f(y)^2 + \sum_i g_i(y)^2 \leq p(y)^2 \ (y \in X)\}$; equivalently $p^*(f + \sum_i t_i g_i) \leq s(|t|)$ for $t \in \mathbb{R}^d$. Theorem 2.1, Theorem 2.2, Corollary 2.3 and Theorem 2.4 hold verbatim for $\mathcal{C}_d$ (support functions on X^d , separation by $(c_0)^d$), with ℓ_2^{d+1} in place of ℓ_2^2 . By [6, Theorem 2.2] there is a dense G_δ set $\Omega \subseteq S_{p^*}$ at which every $\mathcal{C}_d$ is Hausdorff continuous; in particular $\Omega \subseteq \mathcal{R}$. This does not decide density: Hausdorff continuity may fail on a meagre set of first rows, and the remaining sections study what can be recovered there.

3. CERTIFICATES AND THE TRANSPORT THEOREM

Throughout this section $f \in S_{p^*}$ has forced data as in Section 1; block indices are dropped when a single block is involved. For $b \in \ell_1$ we write $\|b/a\|_\infty := \sup_{j \in F} |b_j|/|a_j|$ (when $\text{supp } b \subseteq F$), and $P^\perp$ denotes the orthogonal projection of H onto $e^\perp$.

3.1. Definitions and exact expansions.

Definition 3.1 (Finite certificates). A *finite certificate* at f is a pair $c = (b, \omega)$, $\omega = (\omega_m)_{m \in I}$, such that

- (C1) $b \in \ell_1$, $\text{supp } b \subseteq F$, $\|b/a\|_\infty < \infty$ and $b(\hat{z}) = 0$;
- (C2) each ω_m is finitely supported in Q_m , and $\omega_m = 0$ for all but finitely many m (the *active blocks*).

Put $d_m := \langle D_m w_m, D_m \omega_m \rangle / C_m$ and define the *direction* $g_c := b + \sum_m R_m^*(\omega_m - d_m w_m)$, the coefficients

$$h(b) := \frac{\|P^\perp U^* b\|^2}{\nu}, \quad H_m(\omega_m) := \frac{\|D_m \omega_m\|_2^2 - d_m^2}{C_m},$$

$$H(c) := \max(h(b), \max_m H_m(\omega_m)),$$

the constant $\kappa(c) := \max\{2\beta/\nu, \max_m 2|d_m|M_m/C_m\}$ with $\beta := \|U^* b\|$, and the *coordinatewise radius*

$$r(c) := \min \left\{ \frac{1}{\|b/a\|_\infty}, \frac{\nu}{2\beta}, \min_m \min \left(\min_{k \in \text{supp } \omega_m} \frac{\text{gap}_m(k)}{2|\omega_m(k)|}, \frac{1}{2|d_m|}, \frac{C_m}{2|d_m|M_m} \right) \right\}$$

(with $1/0 = \infty$). Finally

$$\text{Cert}(f) := \{g_c : c \text{ a finite certificate at } f, H(c) \leq 1, g_c \in \mathcal{C}(f)\}.$$

Note that $\|P^\perp D_m \omega_m\|^2 = \|D_m \omega_m\|^2 - d_m^2$, where now $P^\perp$ projects onto $(D_m w_m)^\perp$; so $H_m \geq 0$. The radius is the *coordinatewise* one: each off-peak coordinate is compared with its own gap. (A cruder radius $\min_k \text{gap}(k)/(2 \max_k |\omega(k)|)$ would not suffice for Theorem 4.6 below.)

Lemma 3.2 (Algebra). *Finite certificates form a vector space and $c \mapsto g_c$ is linear; H is convex and $H(\lambda c) = \lambda^2 H(c)$; $r(\lambda c) = r(c)/|\lambda|$, $\kappa(\lambda c) = |\lambda| \kappa(c)$, and $\kappa(c)r(c) \leq 1$. Every direction satisfies $g_c(\xi) = 0$, and $\text{Cert}(f)$ is convex and symmetric.*

Proof. d_m is linear in ω_m , and H is a maximum of positive semidefinite quadratic forms. The inequality $\kappa r \leq 1$ follows from the terms $\nu/(2\beta)$ and $C_m/(2|d_m|M_m)$ of the radius. We have $b(\xi) = q_0 b(\hat{z}) = 0$ and, by Lemma 1.6,

$$\begin{aligned} \frac{\langle \omega_m - d_m w_m, \zeta_m \rangle}{\sigma_m} &= \langle \omega_m, \alpha_m \rangle + \frac{\langle D_m \omega_m, D_m w_m \rangle}{C_m} - d_m \left(\langle w_m, \alpha_m \rangle + \frac{\|D_m w_m\|^2}{C_m} \right) \\ &= 0 + d_m - d_m(M_m + C_m) = 0, \end{aligned}$$

because α_m lives on P_m and ω_m vanishes there. Convexity of $\text{Cert}(f)$ follows from linearity, convexity of H and of $\mathcal{C}(f)$. $\square$

Lemma 3.3 (Exact base identity). *Put $\Psi(h) := \|e + h\| - 1 - \langle e, h \rangle$ for $h \in H$. For $B \in \ell_1$ and $t \in \mathbb{R}$,*

$$(3) \quad q^*(a + tB) = 1 + tB(\hat{z}) + \text{Fl}_B(t) + \sum_{j \notin F} (|tB_j| - z_j tB_j) + \nu \Psi(tU^*B/\nu),$$

where $\text{Fl}_B(t) := \sum_{j \in F} 2(-\text{sign}(a_j)tB_j - |a_j|)_+$; all terms after $tB(\hat{z})$ are nonnegative. Moreover $\Psi(h) = \|h_\perp\|^2/(\|e + h\| + 1 + \langle e, h \rangle)$ with $h_\perp = P^\perp h$, and for $\|h\| \leq \frac{1}{2}$: $\|h_\perp\|^2/(2(1 + \|h\|)) \leq \Psi(h) \leq \|h_\perp\|^2/(2(1 - \|h\|))$. Consequently:

- (a) if $B(\hat{z}) = 0$ and $|t| \|U^*B\| \leq \nu/2$, then $q^*(a + tB) - 1 \geq \frac{t^2}{2} h(B)/(1 + |t| \|U^*B\|/\nu)$;
- (b) if b satisfies (C1) and $|t| \leq \min(1/\|b/a\|_\infty, \nu/(2\beta))$, then $q^*(a + tb) \leq 1 + \frac{t^2}{2} h(b)(1 + 2|t|\beta/\nu)$.

Proof. For real $x \neq 0$ and y : $|x+y| = |x| + \text{sign}(x)y + 2(-\text{sign}(x)y - |x|)_+$. Summing over $j \in F$ (where $z_j = \text{sign } a_j$) gives $\sum_{j \in F} |a_j + tB_j| = \|a\|_1 + \sum_{j \in F} z_j tB_j + \text{Fl}_B(t)$, and $\sum_{j \notin F} |tB_j| = \sum_{j \notin F} z_j tB_j + \sum_{j \notin F} (|tB_j| - z_j tB_j)$. Next $\|U^*a + tU^*B\| = \nu\|e + h\|$ with $h = tU^*B/\nu$ and $\nu\langle e, h \rangle = t\langle U^*B, e \rangle$. Adding and using $\|a\|_1 + \nu = 1$ and $B(\hat{z}) + \langle U^*B, e \rangle = B(\hat{z})$ gives (3). Nonnegativity of Ψ : $\|e + h\| \geq \langle e, e + h \rangle$. The formula for Ψ follows from $\|e + h\|^2 - (1 + \langle e, h \rangle)^2 = \|h\|^2 - \langle e, h \rangle^2$; for $\|h\| \leq \frac{1}{2}$ the denominator lies in $[2(1 - \|h\|), 2(1 + \|h\|)]$. With $h = tU^*B/\nu$ one has $\nu\|h_\perp\|^2 = t^2 h(B)$; (a) follows by dropping the nonnegative terms, and (b) because no flip occurs for $|t| \leq 1/\|b/a\|_\infty$, the sum over $j \notin F$ vanishes, and $1/(1 - x) \leq 1 + 2x$ for $x \leq \frac{1}{2}$. $\square$

Lemma 3.4 (Block expansion). *Fix a block. Let $\omega : \mathbb{N} \rightarrow \mathbb{R}$ vanish on P with $D\omega \in \ell_2$, $d := \langle D\omega, D\omega \rangle/C$, $h_\perp := D\omega - (d/C)D\omega$ (so $\|h_\perp\|^2 = \|D\omega\|^2 - d^2$), $W(\sigma) := (1 - d\sigma)w + \sigma\omega$ and $Y := C + d\sigma M$. Then:*

- (a) $\|DW(\sigma)\|_2 = \sqrt{Y^2 + \sigma^2 \|h_\perp\|^2}$;
- (b) if $1 - d\sigma \geq 0$ and $Y > 0$, then $N(W(\sigma)) \geq 1 + \sigma^2 \|h_\perp\|^2/(\sqrt{Y^2 + \sigma^2 \|h_\perp\|^2} + Y)$;

- (c) if $|d\sigma| \leq \frac{1}{2}$ and $|\sigma\omega(k)| \leq \text{gap}(k)/2$ for all k with $\omega(k) \neq 0$, then $\|W(\sigma)\|_\infty = (1 - d\sigma)M$ and equality holds in (b);
- (d) if $\omega \in c_{00}$ and $|\sigma| \leq \min(\min_k \text{gap}(k)/(2|\omega(k)|), 1/(2|d|), C/(2|d|M))$, then $N(W(\sigma)) \leq 1 + \frac{\sigma^2}{2}H(\omega)(1 + 2|d\sigma|M/C)$.

Proof. (a) Using $1/C - 1 = M/C$, $(1 - d\sigma)Dw + \sigma D\omega = (1 + d\sigma M/C)Dw + \sigma h_\perp$, and $h_\perp \perp Dw$. (b) At a peak k , $|W(\sigma)(k)| = (1 - d\sigma)M$, and $(1 - d\sigma)M + Y = M + C = 1$. (c) Off $\text{supp } \omega$, $|W(k)| = (1 - d\sigma)|w(k)|$; on $\text{supp } \omega$, $|W(k)| \leq (1 - d\sigma)(M - \text{gap}(k)) + \text{gap}(k)/2 \leq (1 - d\sigma)M$. (d) Here $Y \geq C/2$, the bracket in (c) is at most $\sigma^2 \|h_\perp\|^2/(2Y)$, and $C/Y \leq 1/(1 - |d\sigma|M/C) \leq 1 + 2|d\sigma|M/C$. $\square$

Proposition 3.5 (Certificate expansion). *For a finite certificate c at f and $|\sigma| \leq r(c)$,*

$$p^*(f + \sigma g_c) \leq 1 + \frac{\sigma^2}{2}H(c)(1 + \kappa(c)|\sigma|).$$

Proof. $f + \sigma g_c = (a + \sigma b) + \sum_m R_m^* W_m(\sigma)$ with $W_m(\sigma) = (1 - d_m \sigma)w_m + \sigma \omega_m (= w_m$ on inactive blocks). Apply Lemma 1.5, Lemma 3.3(b) and Lemma 3.4(d). $\square$

Lemma 3.6 (Slack and validity window). (a) *For $\rho \in (0, 1)$ and real t , $s(t) - s(\rho t) \geq (1 - \rho^2) \min(t^2, |t|)/3$.* (b) *If $H(c) \leq 1 - \delta$ with $0 < \delta \leq 1$ and $|\sigma| \leq r_*(c, \delta) := \min(r(c), \delta/(2\kappa(c) + 2), \sqrt{\delta})$, then $p^*(f + \sigma g_c) \leq s(\sigma)$.*

Proof. (a) $s(t) - s(\rho t) = (1 - \rho^2)t^2/(s(t) + s(\rho t)) \geq (1 - \rho^2)t^2/(2s(t))$ and $2s(t) \leq 3 \max(1, |t|)$. (b) $(1 - \delta)(1 + \kappa|\sigma|) \leq (1 - \delta)(1 + \delta/2) \leq 1 - \delta/2$, and $1 + \frac{\sigma^2}{2}(1 - \frac{\delta}{2}) \leq 1 + \frac{\sigma^2}{2} - \frac{\sigma^4}{8} \leq s(\sigma)$ since $\sigma^2 \leq \delta$ and $\sqrt{1+x} \geq 1 + \frac{x}{2} - \frac{x^2}{8}$ for $x \geq 0$. $\square$

Proposition 3.7 (Scale criterion). *Let $f, f' \in S_{p^*}$, $g \in \mathcal{C}(f)$, $\rho \in (0, 1)$, and let c' be a finite certificate at f' with $H(c') \leq 1 - \delta$, $0 < \delta \leq 1$. Put $r_* := r_*(c', \delta)$, $\varepsilon := p^*(f' - f)$ and $\eta := p^*(g_{c'} - \rho g)$. If $\varepsilon \leq (1 - \rho^2)r_*^2/6$ and $\eta \leq (1 - \rho^2)r_*/6$, then $g_{c'} \in \text{Cert}(f')$ and $\text{dist}_{p^*}(\rho g, \mathcal{C}(f')) \leq \eta$.*

Proof. For $|t| \leq r_*$ use Lemma 3.6(b) at f' . For $|t| \geq r_*$, $p^*(f' + t g_{c'}) \leq p^*(f + t \rho g) + \varepsilon + |t|\eta \leq s(\rho t) + \varepsilon + |t|\eta$; for $r_* \leq |t| \leq 1$ both ε and $|t|\eta$ are at most $(1 - \rho^2)t^2/6$, and for $|t| \geq 1$ their sum is at most $(1 - \rho^2)|t|/3$; conclude with Lemma 3.6(a). $\square$

Thus the slack alone certifies the scales $|t| \gtrsim \sqrt{\varepsilon/(1 - \rho^2)}$, and the smaller scales must be certified by the local structure of f' ; this is the multi-scale nature of the problem.

3.2. Transport along every sequence.

Theorem 3.8 (Transport). *Let $f_n \rightarrow f$ in S_{p^*} be an arbitrary sequence (norm attaining or not) and let $c = (b, \omega)$ be a finite certificate at f . Put*

$$G_n := \{j \in F : \text{sign } a_n(j) = \text{sign } a_j, |a_n(j)| \geq |a_j|/2\},$$

$$b'_n := b \mathbf{1}_{G_n}, \quad \tau_n := b'_n(\hat{z}_n), \quad b_n := b'_n - \tau_n a_n,$$

and $c_n := (b_n, \omega)$ (the same block coefficients; $d_{n,m}$ is computed at f_n). Then for n large c_n is a finite certificate at f_n , $\|g_{c_n} - g_c\| \rightarrow 0$, $H(c_n) \rightarrow H(c)$, $\kappa(c_n) \rightarrow \kappa(c)$ and $\liminf_n r(c_n) \geq r(c)/2$. If moreover $g_c \in \mathcal{C}(f)$ and $H(c) \leq 1$, then for every $\rho \in (0, 1)$, $\rho g_{c_n} \in \text{Cert}(f_n)$ for all large n .

Proof. Base. By Proposition 1.10, $a_n \rightarrow a$ in ℓ_1 , $\nu_n \rightarrow \nu$, $e_n \rightarrow e$ and $z_n \rightarrow z$ weak*; also $z_n = \text{sign } a_n$ on $\text{supp } a_n$. Every $j \in F$ eventually lies in G_n , so $b'_n \rightarrow b$ in ℓ_1 by dominated convergence. On G_n , $z_n(j) = \text{sign } a_n(j) = \text{sign } a_j = z_j$, hence $\tau_n = \sum_{j \in G_n} b_j z_j + \langle U^* b'_n, e_n \rangle \rightarrow b(z) + \langle U^* b, e \rangle = b(\hat{z}) = 0$, and $b_n \rightarrow b$. Further $\text{supp } b_n \subseteq \text{supp } a_n$, $\|b_n/a_n\|_\infty \leq 2\|b/a\|_\infty + |\tau_n|$, and $b_n(\hat{z}_n) = \tau_n - \tau_n a_n(\hat{z}_n) = 0$. Thus (C1) holds at f_n , $h_n(b_n) \rightarrow h(b)$ and $\|U^* b_n\| \rightarrow \beta$.

Blocks. For the finitely many active m and $k \in \text{supp } \omega_m$, $w_{n,m}(k) \rightarrow w_m(k)$ and $M_{n,m} \rightarrow M_m$, so $M_{n,m} - |w_{n,m}(k)| \rightarrow \text{gap}_m(k) > 0$: eventually $\text{supp } \omega_m \subseteq Q_{n,m}$ and the gaps converge. Also $d_{n,m} \rightarrow d_m$, $C_{n,m} \rightarrow C_m$, $D_m w_{n,m} \rightarrow D_m w_m$, so $H(c_n) \rightarrow H(c)$, $\kappa(c_n) \rightarrow \kappa(c)$ and $\liminf r(c_n) \geq r(c)/2$ (the base ratio is at most halved).

Directions. $g_{c_n} - g_c = (b_n - b) + \sum_m (d_m R_m^* w_m - d_{n,m} R_m^* w_{n,m}) \rightarrow 0$, since $R_m^* w_{n,m} \rightarrow R_m^* w_m$ in norm.

Mates. Let $\delta := (1 - \rho^2)/2$. Then $H(\rho c_n) = \rho^2 H(c_n) \rightarrow \rho^2 H(c) \leq 1 - 2\delta$, so $H(\rho c_n) \leq 1 - \delta$ for large n , and $r_*(\rho c_n, \delta)$ is bounded below by some $r_0 > 0$. Apply Proposition 3.7 with $f' = f_n$, $c' = \rho c_n$ and $g = g_c$: $p^*(f_n - f) \rightarrow 0$ and $p^*(g_{\rho c_n} - \rho g_c) = \rho p^*(g_{c_n} - g_c) \rightarrow 0$. $\square$

Corollary 3.9 (Lower semicontinuity of the certified fibre). *For every sequence $f_n \rightarrow f$ in S_{p^*} ,*

$$\text{cl Cert}(f) \subseteq \text{Li}_n \text{cl Cert}(f_n) \subseteq \text{Li}_n \mathcal{C}(f_n).$$

Consequently: (a) every $g \in \text{cl Cert}(f)$ is recovered along every norm-attaining sequence $f_n \rightarrow f$, finitely many such mates simultaneously, and $(f, g) \in \text{cl NA}((X, p), \ell_2^2)$; (b) if $\mathcal{C}(f) = \text{cl Cert}(f)$, then $f \in \mathcal{R}$ and $\mathcal{C}$ is Hausdorff continuous at f ; (c) if for every f the set $\mathcal{C}(f) \setminus \text{cl Cert}(f)$ is contained in $\text{Li}_n \mathcal{C}(f_n)$ for some norm-attaining sequence $f_n \rightarrow f$, then $\text{NA}((X, p), \ell_2^2)$ is dense.

Proof. By Theorem 3.8, $\rho \text{Cert}(f) \subseteq \text{Li}_n \text{Cert}(f_n)$ for each $\rho < 1$; lower limits are closed. (a) follows from Lemma 1.15 and Corollary 2.3 (applied to finitely many mates). (b) Upper semicontinuity (Theorem 2.1) and $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ give Kuratowski, hence Hausdorff, convergence inside a compact set. (c) Then $\mathcal{C}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ for that sequence (the lower limit is closed and convex); apply Corollary 2.3. $\square$

Corollary 3.10 (Base and block directions). (i) *If ω is finitely supported strictly below the peak of w_m , $d = \langle D_m w_m, D_m \omega \rangle / C_m$, $g = R_m^*(\omega - dw_m) \in \mathcal{C}(f)$ and $H_m(\omega) \leq 1$, then $g \in \text{Cert}(f)$; in particular (f, g) is recovered along every sequence. This is the finite local-certificate theorem quoted from the unavailable note [8] in [6].* (ii) *If b satisfies (C1), then $b/\Lambda \in \text{Cert}(f)$ for every*

$$\Lambda \geq \max(1, \sqrt{2h(b)}, p^*(b)r_0/(s(r_0)-1)), \quad r_0 := \min(r(c), 1/(4\kappa(c)+4), 1/\sqrt{2}),$$

where $c = (b, 0)$.

Proof. (i) is the definition of $\text{Cert}(f)$. (ii) $H(c/\Lambda) = h(b)/\Lambda^2 \leq \frac{1}{2}$, $\kappa(c/\Lambda) \leq \kappa(c)$ and $r(c/\Lambda) \geq r(c)$, so $r_*(c/\Lambda, \frac{1}{2}) \geq r_0$ and Lemma 3.6(b) gives $p^*(f + \sigma b/\Lambda) \leq s(\sigma)$ for $|\sigma| \leq r_0$. For $|\sigma| \geq r_0$, $p^*(f + \sigma b/\Lambda) \leq 1 + |\sigma|p^*(b)/\Lambda \leq s(\sigma)$, since $(s(\sigma)-1)/|\sigma|$ is increasing in $|\sigma|$. $\square$

3.3. Shifted certificates. Since $f = a + L^*w$, one may move a second-order amount of mass between the base and the blocks: $f = \lambda(a + L^*w) - (\lambda - 1)f$ with $\lambda = 1 + \theta\tau^2$.

Definition 3.11 (Shifted certificates). A *shifted certificate* is $c = (b, \omega, \theta)$, where (b, ω) is a finite certificate and $\theta = (\theta_m)$ is a finitely supported real family. Put $v_\theta := \sum_m \theta_m R_m^* w_m \in \ell_1$, $\kappa_q(v) := \sum_{j \notin F} (|v_j| + z_j v_j) \in [0, 2\|v\|_1]$ and

$$H^{\text{sh}}(c) := \max \left(h(b) - \frac{2}{q_0} \sum_m \theta_m \sigma_m + 2\kappa_q(v_\theta), \max_m (H_m(\omega_m) + 2\theta_m) \right).$$

The direction is $g_c := g_{(b, \omega)}$, and $\text{Cert}^{\text{sh}}(f) := \{g_c \in \mathcal{C}(f) : H^{\text{sh}}(c) \leq 1\}$.

Note $v_\theta(\hat{z}) = \sum_m \theta_m \langle w_m, \zeta_m \rangle / q_0 = \sum_m \theta_m \sigma_m / q_0$. The associated decomposition is

$$(4) \quad \begin{aligned} A(\tau) &:= a + \tau b - \tau^2 v_\theta, & W_m(\tau) &:= (1 + \theta_m \tau^2) w_m + \tau(\omega_m - d_m w_m), \\ A(\tau) + L^* W(\tau) &= f + \tau g_c. \end{aligned}$$

For $\rho > 0$ put $\rho.c := (\rho b, \rho \omega, \rho^2 \theta)$; then $g_{\rho.c} = \rho g_c$ and $H^{\text{sh}}(\rho.c) = \rho^2 H^{\text{sh}}(c)$. H^{sh} is jointly convex, so $\text{Cert}^{\text{sh}}(f)$ is convex, symmetric and contains $\text{Cert}(f)$ ($\theta = 0$).

Proposition 3.12 (Shifted expansion). *For a shifted certificate c at f there is $\varepsilon_c(\tau) \rightarrow 0$ with $p^*(f + \tau g_c) \leq 1 + \frac{\tau^2}{2}(H^{\text{sh}}(c) + \varepsilon_c(\tau))$.*

Proof. Blocks. $W_m(\tau) = (1 + \theta_m \tau^2)[w_m + \tau'(\omega_m - d_m w_m)]$ with $\tau' = \tau/(1 + \theta_m \tau^2)$, so Lemma 3.4(d) gives $N_m(W_m(\tau)) \leq 1 + \tau^2(\theta_m + H_m/2) + O(|\tau|^3)$. *Base.* Apply (3) with $t = \tau$, $B = b - \tau v_\theta$: $\tau B(\hat{z}) = -\tau^2 v_\theta(\hat{z})$; off F , $|\tau B_j| - z_j \tau B_j = \tau^2(|v_j| + z_j v_j)$; for $|\tau| \leq 1/(2\|b/a\|_\infty)$ each flip term is at most $2(\tau^2|v_j| - |a_j|/2)_+$, so $\text{Fl} \leq 2\tau^2 \sum_{j \in F, |v_j| > |a_j|/(2\tau^2)} |v_j| = o(\tau^2)$; and $\nu \Psi(\tau U^* B / \nu) = \frac{\tau^2}{2}(h(b) + O(\tau))$. Hence $q^*(A(\tau)) \leq 1 + \frac{\tau^2}{2}(h(b) - 2v_\theta(\hat{z}) + 2\kappa_q(v_\theta)) + o(\tau^2)$. Conclude with Lemma 1.5. $\square$

Theorem 3.13 (Shifted transport). *Let $f_n \rightarrow f$ in S_{p^*} be arbitrary, $c = (b, \omega, \theta)$ a shifted certificate at f , and $c_n := (b_n, \omega, \theta)$ with b_n as in Theorem 3.8. For every $\eta > 0$ there are $\tau_\eta > 0$ and n_η with $p^*(f_n + \tau g_{c_n}) \leq 1 + \frac{\tau^2}{2}(H^{\text{sh}}(c) + \eta)$ for $n \geq n_\eta$, $|\tau| \leq \tau_\eta$; moreover $\limsup_n H_n^{\text{sh}}(c_n) \leq H^{\text{sh}}(c)$ (computed at f_n). Consequently, if $g_c \in \mathcal{C}(f)$ and $H^{\text{sh}}(c) \leq 1$, then $\rho g_{c_n} \in \text{Cert}^{\text{sh}}(f_n)$ for every $\rho < 1$ and n large, and*

$$\text{cl Cert}^{\text{sh}}(f) \subseteq \text{Li cl Cert}^{\text{sh}}(f_n) \subseteq \text{Li } \mathcal{C}(f_n) \quad \text{for every sequence } f_n \rightarrow f.$$

Proof. The blocks are treated as in Proposition 3.12, with the uniform bounds of Theorem 3.8. For the base let $v_n := \sum_m \theta_m R_m^* w_{n,m} \rightarrow v_\theta$ in ℓ_1 and apply (3) at f_n with $B = b_n - \tau v_n$. For $|\tau| \leq 1/(2\|b_n/a_n\|_\infty)$ and $|\tau \tau_n| \leq 1$ one checks coordinatewise: for $j \in G_n$ the flip term is at most $2\tau^2|v_{n,j}|1[|v_{n,j}| > |a_j|/(4\tau^2)]$; for $j \in \text{supp } a_n \setminus G_n$ one has $b_n(j) = -\tau_n a_n(j)$ and the flip term is at most $\tau^2(|v_{n,j}| + z_n(j)v_{n,j})$; off $\text{supp } a_n$ the kink term equals $\tau^2(|v_{n,j}| + z_n(j)v_{n,j})$. Hence the flip and kink terms are at most τ^2 times

$$2 \sum_{j \in F} |v_{n,j}| 1[|v_{n,j}| > \frac{|a_j|}{4\tau^2}] + \sum_{j \notin F} (|v_{n,j}| + z_n(j)v_{n,j}) + 2 \sum_{j \in F \setminus G_n} |v_{n,j}|.$$

As $n \rightarrow \infty$ and $\tau \rightarrow 0$ the first and third sums tend to 0 (compare v_n with v_θ in ℓ_1 , use dominated convergence and that each $j \in F$ eventually lies in G_n), and the second tends to $\kappa_q(v_\theta)$ ($z_n \rightarrow z$ coordinatewise, $|z_n| \leq 1$). Also $v_n(\hat{z}_n) \rightarrow$

$v_\theta(\hat{z})$, $\sigma_{n,m} \rightarrow \sigma_m$, $q_0^{(n)} \rightarrow q_0$, and the Hilbert term is $\frac{\tau^2}{2}(h_n(b_n) + O(\tau))$ with $h_n(b_n) \rightarrow h(b)$. This gives the uniform estimate, and the same computation without τ gives the lim sup of $H_n^{\text{sh}}(c_n)$. For the mates, fix $\rho < 1$ and $\eta \leq (1 - \rho)/\rho$; for $|\tau| \leq t_1 := \min(\tau_\eta/\rho, 2\sqrt{1 - \rho})$ we get $p^*(f_n + \tau\rho g_{c_n}) \leq 1 + \rho\tau^2/2 \leq s(\tau)$, and for $|\tau| \geq t_1$ we use the slack (Lemma 3.6(a)) with $p^*(f_n - f) \rightarrow 0$ and $p^*(g_{c_n} - g_c) \rightarrow 0$. Finally $H_n^{\text{sh}}(\rho.c_n) = \rho^2 H_n^{\text{sh}}(c_n) \leq 1$ for large n . $\square$

Remark 3.14. Shifts relax the condition $H \leq 1$ to $H^{\text{sh}} \leq 1$. Whether $\text{cl Cert}^{\text{sh}}(f)$ is strictly larger than $\text{cl Cert}(f)$ for some f in Martín's space is *open*: no mate with $H > 1 \geq H^{\text{sh}}$ outside $\text{cl Cert}(f)$ has been exhibited (finite-dimensional models suggest that shifts help, but they cannot distinguish $\text{Cert}(f)$ from its closure). Section 5 identifies the sharp second-order invariant at tame supports, Γ_w , and one checks that $\min_\theta H^{\text{sh}}(b, \omega, \theta) \geq \Gamma_w$ (Remark 5.10).

3.4. Several rows.

Theorem 3.15 (Joint fibres). *Let $c_1, \dots, c_d$ be finite certificates at f with $G := (g_{c_1}, \dots, g_{c_d}) \in \mathcal{C}_d(f)$ and $\sup_{\theta \in S^{d-1}} H(\sum_i \theta_i c_i) \leq 1$. Then for every sequence $f_n \rightarrow f$ in S_{p^*} and every $\rho < 1$ the transported tuple $\rho G_n := (\rho g_{c_{1,n}}, \dots, \rho g_{c_{d,n}})$ lies in $\mathcal{C}_d(f_n)$ for n large, and $G_n \rightarrow G$. In particular, if the f_n attain their norms, then $(f, \rho G)$ is a limit of norm-attaining operators into ℓ_2^{d+1} .*

Proof. The construction of Theorem 3.8 is linear in c (the sets G_n do not depend on b), so $c(\theta) := \sum_i \theta_i c_i$ is transported to $c_n(\theta) = \sum_i \theta_i c_{i,n}$. On S^{d-1} the coefficients $H(c_n(\theta))$ converge to $H(c(\theta))$ uniformly (they are maxima of quadratic forms in θ with convergent coefficients), $\kappa(c_n(\theta))$ is bounded, and $r(c_n(\theta)) \geq r_0 > 0$ uniformly for n large, because only the finitely many coordinates of $\bigcup_i \text{supp } \omega_i$ and the bounded quantities $\|b_{i,n}/a_n\|_\infty$, $\|U^* b_{i,n}\|$, $|d_{i,n,m}|$ enter. Now run the proof of Proposition 3.7 along each ray $t = |t|\theta$: for $|t| \leq r_0$ use Lemma 3.6(b) for $\rho c_n(\theta)$; for $|t| \geq r_0$ use $p^*(f + \rho \sum_i t_i g_{c_i}) \leq s(\rho|t|)$ and the slack in $|t|$. If f_n attains its norm at x_n , then each row of ρG_n vanishes at x_n and $(f_n, \rho G_n)$ attains its norm there. $\square$

4. AVERAGING OVER SCALES

4.1. The averaging theorem.

Theorem 4.1 (Weighted averaging). *Let $g \in \mathcal{C}(f)$ and $\rho \in (0, 1)$. Let $c_1, \dots, c_n$ be finite certificates at f and $\mu_1, \dots, \mu_n \geq 0$ weights with $\sum_i \mu_i = 1$; put $r_i := r(c_i)$ and $e_i := p^*(g - g_{c_i})$. Let $s_0 \in (0, \min(1, \sqrt{1 - \rho^2})]$ and $\eta_0, \kappa_0 \geq 0$ satisfy $\rho^2(1 + \eta_0)(1 + \kappa_0) \leq (1 + \rho^2)/2$, and assume*

- (i) $H(c_i) \leq 1 + \eta_0$ and $\kappa(c_i) \min(r_i, s_0) \leq \kappa_0$ for all i ;
- (ii) $\sum_{i: r_i < \rho s} \mu_i e_i \leq (1 - \rho^2)s/(12\rho)$ for every $s \in (0, s_0]$;
- (iii) $\sum_i \mu_i e_i \leq (1 - \rho^2)s_0/(3\rho)$.

Then $c := \sum_i \mu_i c_i$ satisfies $\rho g_c \in \text{Cert}(f)$, and $p^(\rho g_c - \rho g) \leq \rho \sum_i \mu_i e_i$. Consequently, if for every $\rho < 1$ and $\varepsilon > 0$ such families exist with $\sum_i \mu_i e_i \leq \varepsilon$, then $g \in \text{cl Cert}(f)$.*

Proof. By linearity $g_c = \sum_i \mu_i g_{c_i}$, and by convexity of p^* , $p^*(f + s\rho g_c) \leq \sum_i \mu_i p^*(f + s\rho g_{c_i})$ for real s . We use two bounds for each i . (A) If $\rho|s| \leq r_i$ and $|s| \leq s_0$, Proposition 3.5 and (i) give $p^*(f + s\rho g_{c_i}) \leq 1 + \frac{\rho^2 s^2}{2}(1 + \eta_0)(1 + \kappa_0) \leq 1 + \frac{s^2}{4}(1 + \rho^2)$,

since $\kappa(c_i)\rho|s| \leq \kappa(c_i)\min(r_i, s_0)$. (B) Always, since $g \in \mathcal{C}(f)$, $p^*(f + s\rho g_{c_i}) \leq p^*(f + s\rho g) + \rho|s|e_i \leq s(\rho s) + \rho|s|e_i$.

Case $|s| \leq s_0$. Use (A) for $r_i \geq \rho|s|$ and (B) otherwise. By (ii),

$$p^*(f + s\rho g_c) \leq \max\left(1 + \frac{s^2}{4}(1 + \rho^2), s(\rho s)\right) + \frac{(1 - \rho^2)s^2}{12}.$$

Now $1 + \frac{s^2}{4}(1 + \rho^2) + \frac{(1 - \rho^2)s^2}{12} = 1 + \frac{s^2}{6}(2 + \rho^2) = 1 + \frac{s^2}{2} - \frac{(1 - \rho^2)s^2}{6} \leq 1 + \frac{s^2}{2} - \frac{s^4}{8} \leq s(s)$ because $s^2 \leq 1 - \rho^2$; and $s(\rho s) + (1 - \rho^2)s^2/12 \leq s(s)$ by Lemma 3.6(a).

Case $|s| \geq s_0$. Use (B) for every i and (iii): $p^*(f + s\rho g_c) \leq s(\rho s) + (1 - \rho^2)s_0|s|/3 \leq s(s)$, because $\min(s^2, |s|) \geq s_0|s|$.

Thus $\rho g_c \in \mathcal{C}(f)$. By convexity of H , $H(\rho c) \leq \rho^2 \max_i H(c_i) \leq \rho^2(1 + \eta_0) \leq 1$, so $\rho g_c \in \text{Cert}(f)$. The distance estimate is the triangle inequality. $\square$

The mechanism is that at each scale s only the certificates that are too fine for s contribute an error, of order s each, and these errors are summable against the weights.

Corollary 4.2 (Ladder form). *Let $g \in \mathcal{C}(f)$. Suppose there are $c_0 > 0$, $\kappa_1, K < \infty$, $t_* > 0$ and a function $\eta(t) \rightarrow 0$ ($t \rightarrow 0^+$) such that for every $t \in (0, t_*]$ there is a finite certificate c_t at f with*

$$H(c_t) \leq 1 + \eta(t), \quad r(c_t) \geq c_0 t, \quad \kappa(c_t) \leq \kappa_1, \quad p^*(g - g_{c_t}) \leq Kt.$$

Then $g \in \text{cl Cert}(f)$; hence g is recovered along every sequence $f_n \rightarrow f$ (Corollary 3.9). It suffices that the hypotheses hold along a geometric sequence of scales $t_1 2^{1-i}$.

Proof. Fix $\rho < 1$; choose $\eta_0, \kappa_0 > 0$ with $\rho^2(1 + \eta_0)(1 + \kappa_0) \leq (1 + \rho^2)/2$, then $s_0 \in (0, \min(1, \sqrt{1 - \rho^2})]$ with $\kappa_1 s_0 \leq \kappa_0$, an integer $n \geq 24\rho^2 K/(c_0(1 - \rho^2))$, and $t_1 \in (0, t_*]$ with $\eta(t) \leq \eta_0$ for $t \leq t_1$ and $2\rho K t_1/n \leq (1 - \rho^2)s_0/3$. Put $t_i := t_1 2^{1-i}$, $c_i := c_{t_i}$ and $\mu_i := 1/n$ ($i \leq n$). Condition (i) holds. For (ii), $r_i < \rho s$ implies $c_0 t_i < \rho s$, and the t_i with this property sum to less than $2\rho s/c_0$, so the left side of (ii) is at most $(K/n)(2\rho s/c_0) \leq (1 - \rho^2)s/(12\rho)$. For (iii), $\sum_i \mu_i K t_i \leq 2K t_1/n$. Theorem 4.1 gives $\rho g_c \in \text{Cert}(f)$ within $2\rho K t_1/n$ of ρg ; let $t_1 \rightarrow 0$ and then $\rho \rightarrow 1$. $\square$

Corollary 4.3 (Decomposition form). *Let $g \in \mathcal{C}(f)$. Suppose that for every small $t > 0$ there are a finite certificate $c_t = (b_t, \omega_t)$ with $r(c_t) \geq c_0 t$ and $\kappa(c_t) \leq \kappa_1$, and $e_t := g - g_{c_t}$ with $p^*(e_t) \leq Kt$, such that for $\tau = t$ or for $\tau = -t$ the decomposition*

$$f + \tau g = (a + \tau b_t + \tau e_t) + \sum_m R_m^*((1 - d_{t,m}\tau)w_m + \tau \omega_{t,m})$$

has all component norms at most $s(\tau) + o(\tau^2)$. Assume moreover that $t^2/C_m \rightarrow 0$ uniformly over the active blocks of c_t (this is automatic if I is finite). Then $H(c_t) \leq 1 + o(1)$, and Corollary 4.2 gives $g \in \text{cl Cert}(f)$.

Proof. Blocks. Since $\kappa(c_t) \leq \kappa_1$, $|d_{t,m}| \leq \kappa_1 C_m/(2M_m)$, so $Y = C_m + d_{t,m}\tau M_m = C_m(1 + O(t))$ and $1 - d_{t,m}\tau > 0$ for small t . Put $X := |\tau||h_\perp|$. Lemma 3.4(b) gives $\sqrt{Y^2 + X^2} - Y \leq \delta := s(\tau) - 1 + o(\tau^2)$, hence $X^2 \leq \delta(2Y + \delta)$ and

$$H_m(\omega_{t,m}) = \frac{\|h_\perp\|^2}{C_m} \leq \frac{\delta}{\tau^2} \left(\frac{2Y}{C_m} + \frac{\delta}{C_m} \right) = 1 + o(1) + O(\tau^2/C_m).$$

Base. $B := b_t + e_t$ satisfies $B(\hat{z}) = 0$, because $g(\xi) = 0$ and $g_{e_t}(\xi) = 0$ (Lemma 3.2). Also $\|U^*b_t\| \leq \kappa_1\nu/2$ and $\|e_t\|_1 \leq q^*(e_t) \leq (1 + \|L\|)p^*(e_t) \leq (1 + \|L\|)Kt$. Lemma 3.3(a) gives $h(B) \leq 1 + o(1)$, and since $\sqrt{h}$ is a seminorm and $h(e_t) \leq \|U\|^2\|e_t\|_1^2/\nu = O(t^2)$, also $h(b_t) \leq 1 + o(1)$. $\square$

Remark 4.4. For $I = \mathbb{N}$ the extra hypothesis in Corollary 4.3 cannot simply be dropped: $C_m \leq \|\Phi_m\|_2 \leq 2^{-m}$, and a block with $C_m \ll \tau^2$ can remain admissible at scale τ with $\|h_\perp\| \approx \tau/2 + C_m/\tau$, i.e. with $H_m \approx \tau^2/(4C_m) \gg 1$. Corollary 4.2 itself has no such restriction.

4.2. Weighted directions. For a block m , a function $\omega : \mathbb{N} \rightarrow \mathbb{R}$ vanishing on P_m and $\sigma > 0$, put

$$E_m(\sigma; \omega) := \{k \in Q_m : \sigma|\omega(k)| > \text{gap}_m(k)/2\}, \quad T_m(\sigma; \omega) := \sum_{k \in E_m(\sigma; \omega)} \lambda_{k,m}|\omega(k)|.$$

Lemma 4.5 (Box tails). $T_m(\sigma; \omega) = o(\sigma)$ as $\sigma \rightarrow 0$ in each of the following cases: (a) $\sum_k \lambda_{k,m}\omega(k)^2/\text{gap}_m(k) < \infty$; (b) ω is bounded and $|\omega(k)| \leq K\sqrt{\text{gap}_m(k)}$ whenever $\text{gap}_m(k) \leq \gamma_0$, for some $K, \gamma_0 > 0$; (c) $\text{gap}_m \geq M_m/2$ on $\text{supp } \omega$ and $\sum_k \lambda_{k,m}\omega(k)^2 < \infty$ (the weighted directions of [6, Theorem 3.1]).

Proof. (a) On $E_m(\sigma)$, $\lambda_k|\omega(k)| < 2\sigma\lambda_k\omega(k)^2/\text{gap}(k)$, and the sets $E_m(\sigma)$ decrease to $\emptyset$ as $\sigma \downarrow 0$; use dominated convergence. (b) If $\sigma\|\omega\|_\infty < \gamma_0/2$, then $k \in E_m(\sigma)$ forces $\text{gap}(k) \leq \gamma_0$, hence $\text{gap}(k) < 2\sigma K\sqrt{\text{gap}(k)}$, i.e. $\text{gap}(k) < 4K^2\sigma^2$ and $|\omega(k)| < 2K^2\sigma$; so $T_m(\sigma) \leq 2K^2\sigma \sum_{\text{gap}(k) < 4K^2\sigma^2} \lambda_k = o(\sigma)$. (c) is a case of (a). $\square$

Theorem 4.6 (Weighted directions). Let $b \in \ell_1$ with $\text{supp } b \subseteq F$, $b(\hat{z}) = 0$ and $\sum_{j \in F, |b_j| > |a_j|/(2t)} |b_j| = O(t)$ (e.g. $\sum_{j \in F} b_j^2/|a_j| < \infty$). For finitely many blocks m let $\omega_m : \mathbb{N} \rightarrow \mathbb{R}$ vanish on P_m , with $\sum_k \lambda_{k,m}|\omega_m(k)| < \infty$ and $T_m(\sigma; \omega_m) = O(\sigma)$. Put $d_m := \langle D_m\omega_m, D_m\omega_m \rangle / C_m$, $g := b + \sum_m R_m^*(\omega_m - d_m\omega_m)$ and $H := \max(h(b), \max_m H_m(\omega_m))$. If $g \in \mathcal{C}(f)$ and $H \leq 1$, then $g \in \text{cl Cert}(f)$; in particular g is recovered along every sequence $f_n \rightarrow f$.

Proof. Since $\Phi_m \leq \lambda_{\cdot,m}$, $D_m\omega_m \in \ell_1 \subseteq \ell_2$. For small $t > 0$ choose $J_t \rightarrow \infty$ so large that the tails of b and of $\lambda_{\cdot,m}\omega_m$ beyond J_t have ℓ_1 -norm at most t , and put

$$G_t := F \cap [1, J_t] \cap \{|b_j| \leq |a_j|/(2t)\}, \quad b_t := b1_{G_t} - \varkappa_t a, \quad \varkappa_t := (b1_{G_t})(\hat{z}),$$

$$\omega_{m,t} := \omega_m 1_{\{k \leq J_t\} \setminus E_m(t; \omega_m)}.$$

Then $c_t := (b_t, (\omega_{m,t}))$ is a finite certificate. Since $|\varkappa_t| = |(b1_{G_t})(\hat{z})| \leq q^*(b1_{F \setminus G_t}) \leq (1 + \|U\|)\|b1_{F \setminus G_t}\|_1 = O(t)$, we get $\|b_t/a\|_\infty \leq 1/(2t) + O(t)$, so the base part of $r(c_t)$ is at least t for small t ; the block coordinates kept satisfy $t|\omega_{m,t}(k)| \leq \text{gap}_m(k)/2$, and the remaining terms of the radius are bounded below (the $d_{m,t}$ and $\|U^*b_t\|$ are bounded). Hence $r(c_t) \geq t$ and $\kappa(c_t)$ is bounded. The remainder is

$$g - g_{c_t} = (b - b_t) + \sum_m R_m^*((\omega_m - \omega_{m,t}) - (d_m - d_{m,t})\omega_m),$$

with $\|b - b_t\|_1 \leq \|b1_{F \setminus G_t}\|_1 + |\varkappa_t| = O(t)$, $\sum_k \lambda_{k,m}|\omega_m - \omega_{m,t}|(k) \leq T_m(t; \omega_m) + t = O(t)$ and $\|d_m - d_{m,t}\|_2 \leq \|D_m(\omega_m - \omega_{m,t})\|_2 = O(t)$. So $p^*(g - g_{c_t}) \leq Kt$. Finally $b_t \rightarrow b$ in ℓ_1 and $D_m\omega_{m,t} \rightarrow D_m\omega_m$ in ℓ_2 , so $H(c_t) \rightarrow H \leq 1$. Corollary 4.2 applies. $\square$

By Lemma 4.5(c), Theorem 4.6 contains [6, Theorem 3.1] (and removes its dependence on [8]); near-peak supports are allowed as long as the box tails are $O(\sigma)$. In particular the non-attaining examples of [6, Section 3], which admit no bounded first-order lift, are recovered along every sequence.

4.3. Locally split mates.

Definition 4.7. A mate $g \in \mathcal{C}(f)$ is *locally split* if there are $\tau > 0$, $b \in X^*$ and $v \in V^*$ with $g = b + L^*v$ and, for $|t| \leq \tau$,

$$q^*(a + tb) \leq s(t) \quad \text{and} \quad \|w + tv\|_{V^*} \leq s(t).$$

Equivalently, the linear path $f + tg = (a + tb) + L^*(w + tv)$ is admissible at level $s(t)$ near $t = 0$. Every finite certificate direction with $H \leq 1$ is locally split (Proposition 3.5); the converse fails, since b and v may have infinite supports.

Lemma 4.8 (Base truncation with exact first-order correction). *Let $b \in \ell_1$ satisfy $q^*(a + tb) \leq s(t)$ for $|t| \leq \tau$. Then $\text{supp } b \subseteq F$ and $b(\hat{z}) = 0$. Put $\tau_a := \frac{1}{4} \min(\tau, 1, \nu/(2\|U^*b\| + 1))$. For every $\delta > 0$ there is N such that $b'' := P_N b + \beta_N a$, with P_N the coordinate projection onto $[1, N]$ and $\beta_N := ((I - P_N)b)(\hat{z})$, satisfies $b''(\hat{z}) = 0$, $\|b'' - b\|_1 < \delta$, and $q^*(a + tb'') \leq s(t) + \delta t^2$ for $|t| \leq \tau_a$.*

Proof. First order. The function $\varphi(t) = q^*(a + tb)$ is convex with $\varphi(0) = 1$ and $\varphi(t) \leq 1 + t^2/2$ on $[-\tau, \tau]$, so both one-sided derivatives at 0 vanish. The right derivative is $\sum_{j \in F} \text{sign}(a_j)b_j + \sum_{j \notin F} |b_j| + \langle U^*b, e \rangle$ and the left derivative is $\sum_{j \in F} \text{sign}(a_j)b_j - \sum_{j \notin F} |b_j| + \langle U^*b, e \rangle$; hence $\sum_{j \notin F} |b_j| = 0$ and $b(\hat{z}) = 0$.

Truncation. Let $r := (I - P_N)b$, $\varepsilon_N := \|U^*r\|$, and note $\beta_N = r(\hat{z}) = \sum_{j > N} z_j r_j + \langle U^*r, e \rangle \rightarrow 0$. For $|s| \leq 2\tau_a$ and $j > N$ in F , $|a_j| \leq |a_j + sr_j| - sz_j r_j$; hence $\|a + sP_N b\|_1 \leq \|a + sb\|_1 - s \sum_{j > N} z_j r_j$. For the Hilbert part we use the elementary inequality

$$(5) \quad \|y - x\| \leq \|y\| - \langle y, x \rangle / \|y\| + \|x\|^2 / (2\|y\|) \quad (y \neq 0),$$

which follows from $(\langle y, x \rangle / \|y\|^2 - \|x\|^2 / (2\|y\|^2))^2 \geq 0$. With $y_s := U^*(a + sb)$ (so $\|y_s\| \geq \frac{3}{4}\nu$ and $\|y_s\| \|y_s\| - e\| \leq 2|s| \|U^*b\| / \nu$) and $x = sU^*r$, $\|U^*(a + sP_N b)\| \leq \|y_s\| - s\langle e, U^*r \rangle + C_2 s^2 \varepsilon_N$, with C_2 depending only on ν , $\|U^*b\|$. Adding,

$$q^*(a + sP_N b) \leq q^*(a + sb) - s\beta_N + C_2 s^2 \varepsilon_N.$$

Now $a + tb'' = (1 + t\beta_N)(a + sP_N b)$ with $s = t/(1 + t\beta_N)$; for N large ($|\beta_N| \leq 1$) and $|t| \leq \tau_a$ we have $|s| \leq 2\tau_a$, and

$$q^*(a + tb'') \leq (1 + t\beta_N)s(s) - t\beta_N + C_3 t^2 \varepsilon_N = 1 + F_t(\beta_N) + C_3 t^2 \varepsilon_N,$$

where $F_t(\beta) := \sqrt{(1 + t\beta)^2 + t^2} - (1 + t\beta)$. Since $F_t(0) = s(t) - 1$ and $|\partial_\beta F_t| = |t|(1 + t\beta) / \sqrt{(1 + t\beta)^2 + t^2} - 1| \leq 2|t|^3$ for $|t\beta| \leq \frac{1}{2}$, we get $q^*(a + tb'') \leq s(t) + 2|\beta_N||t|^3 + C_3 t^2 \varepsilon_N \leq s(t) + \delta t^2$ for N large. Finally $\|b'' - b\|_1 \leq \|r\|_1 + |\beta_N| \|a\|_1 \rightarrow 0$ and $b''(\hat{z}) = b(\hat{z}) - r(\hat{z}) + \beta_N = 0$. $\square$

Lemma 4.9 (Block truncation). *Fix a block (drop m) and let $v \in \ell_\infty$ satisfy $N(w + tv) \leq s(t)$ for $|t| \leq \tau$. Put $c := \langle Dw, Dv \rangle / C$, $d := c/M$ and $\omega := v + dw$. Then $\omega = 0$ on P , $|\text{sign}(w(k))v(k) + c| \leq \sqrt{2 \text{gap}(k)}$ whenever $w(k) \neq 0$ and $\text{gap}(k) \leq \tau^2/2$, and $d = \langle Dw, D\omega \rangle / C$. Moreover there is $\tau_3 > 0$ such that for every $\delta > 0$ there is a finitely supported ω'' , vanishing on P , with $\min_{\text{supp } \omega''} \text{gap} > 0$ and $\langle Dw, D\omega'' \rangle = \langle Dw, D\omega \rangle$, such that $v'' := \omega'' - dw$ satisfies $\|R^*(v'' - v)\|_1 < \delta$ and $N(w + tv'') \leq N(w + tv) + \delta t^2$ for $|t| \leq \tau_3$.*

Proof. Structure. $\chi(t) := \|D(w+tv)\|$ is convex and differentiable at 0 with $\chi'(0) = c$, so $\chi(t) \geq C + tc$, and $\|w+tv\|_\infty \leq s(t) - \chi(t) \leq M - tc + t^2/2$ for $|t| \leq \tau$. For k with $w(k) \neq 0$ and $\varsigma_k := \text{sign } w(k)$ this gives $|w(k)| + t\varsigma_k v(k) \leq M - tc + t^2/2$, i.e. $t(\varsigma_k v(k) + c) \leq \text{gap}(k) + t^2/2$ for $|t| \leq \tau$. For a peak ($\text{gap} = 0$) both signs of t give $\varsigma_k v(k) = -c$; for $\text{gap}(k) \leq \tau^2/2$ take $t = \pm\sqrt{2\text{gap}(k)}$. Then $\omega(k) = \varsigma_k(\varsigma_k v(k) + c) = 0$ on P , and off P with $w(k) \neq 0$, $\omega(k) = \varsigma_k((\varsigma_k v(k) + c) - c\text{gap}(k)/M)$, which tends to 0 along every sequence with $\text{gap}(k) \rightarrow 0$; ω is bounded. Finally $\langle Dw, D\omega \rangle = cC + dC^2 = cC(M+C)/M = dC$.

Truncation. If $w(k) = 0$ for every $k \in Q$, then $\langle Dw, D\omega \rangle = 0$, $d = c = 0$, and we put $k_0 := \text{none}$, $\varkappa := 0$. Otherwise fix $k_0 \in Q$ with $w(k_0) \neq 0$, $\gamma_0 := \text{gap}(k_0)$. For $\gamma \in (0, \gamma_0)$ and $K \geq k_0$ let $S := \{k \leq K : \text{gap}(k) \geq \gamma\}$ (finite, contains k_0) and

$$\omega'' := \omega 1_S + \varkappa e_{k_0}, \quad \varkappa := \langle Dw, D(\omega - \omega 1_S) \rangle / (\Phi(k_0)^2 w(k_0)),$$

so that $\langle Dw, D\omega'' \rangle = \langle Dw, D\omega \rangle$. As $\gamma \rightarrow 0$ and $K \rightarrow \infty$, $\omega - \omega 1_S \rightarrow 0$ pointwise and boundedly ($\omega = 0$ on P), hence $\Delta := D(\omega'' - \omega) \rightarrow 0$ in ℓ_2 , $\varkappa \rightarrow 0$ and $\|R^*(\omega'' - \omega)\|_1 \leq \sum_k \lambda_k |\omega''(k) - \omega(k)| \rightarrow 0$.

Sup part. Let $W := w + tv = (1-dt)w + tw$ and $W'' := w + tv'' = (1-dt)w + t\omega''$. They agree on $S \setminus \{k_0\}$. For $k \notin S \cup \{k_0\}$, $|W''(k)| = |1-dt||w(k)| \leq |1-dt|M = |W(k_*)|$ for any peak k_* ($\omega(k_*) = 0$), so $|W''(k)| \leq \|W\|_\infty$. At k_0 , $|W''(k_0)| \leq |1-dt|(M - \gamma_0) + |t|(\|\omega\|_\infty + 1) \leq |1-dt|M$ for $|t| \leq t_0 := \gamma_0/(4(|d|M + \|\omega\|_\infty + 2))$ (once $|\varkappa| \leq 1$). Hence $\|W''\|_\infty \leq \|W\|_\infty$ for $|t| \leq t_0$.

Hilbert part. Put $A_t := DW$. Since $\langle Dw, \Delta \rangle = 0$, $\langle A_t, \Delta \rangle = t\langle D\omega, \Delta \rangle$ and

$$\begin{aligned} \|A_t + t\Delta\|^2 &= \|A_t\|^2 + 2t^2\langle D\omega, \Delta \rangle + t^2\|\Delta\|^2, \quad \text{so} \\ \|A_t + t\Delta\| &\leq \|A_t\| + \frac{2t^2(\|D\omega\|\|\Delta\| + \|\Delta\|^2)}{C} \end{aligned}$$

for $|t| \leq t_1 := C/(2\|D(\omega - dw)\| + 2)$ (then $\|A_t\| \geq C/2$). Adding, $N(W'') \leq N(W) + o(1)t^2$ on $|t| \leq \tau_3 := \min(\tau, t_0, t_1)$, where $o(1) \rightarrow 0$ as $\gamma \rightarrow 0$, $K \rightarrow \infty$. $\square$

Theorem 4.10 (Locally split mates). *Let I be any nonempty block set. Every locally split mate $g \in \mathcal{C}(f)$ lies in $\text{cl Cert}(f)$; in particular $(f, g) \in \text{cl NA}((c_0, p), \ell_2^2)$ and g is recovered along every sequence $f_n \rightarrow f$ in S_{p^*} .*

Proof. Let $g = b + L^*v$ be as in Definition 4.7. Fix $\rho \in (0, 1)$ and $\delta > 0$. Choose a finite set I_0 of blocks with $\|v\| \sum_{m \notin I_0} m2^{-m} < \delta$ and replace v_m by 0 for $m \notin I_0$; this keeps the admissibility ($N_m(w_m) = 1$) and moves g by at most $(1 + \|U\|)\delta$ in p^* . Apply Lemma 4.8 to b and Lemma 4.9 to each v_m , $m \in I_0$: we obtain a finite certificate $c'' = (b'', (\omega''_m)_{m \in I_0})$ (its d -coefficients are the d_m of the lemma) with $\|g_{c''} - g\|_{p^*} \leq \delta'$, $\delta' \rightarrow 0$ as $\delta \rightarrow 0$, and a range $|t| \leq \tau_5$, independent of δ , on which

$$q^*(a + tb'') \leq s(t) + \delta t^2, \quad N_m(w_m + t(\omega''_m - d_m w_m)) \leq s(t) + \delta t^2.$$

Coefficient. Lemma 3.3(a) gives $\frac{t}{2}h(b'')/(1 + |t|\|U^*b''\|/\nu) \leq s(t) - 1 + \delta t^2$; dividing by t^2 and letting $t \rightarrow 0$, $h(b'') \leq 1 + 2\delta$. Similarly Lemma 3.4(b) gives $H_m(\omega''_m) \leq 1 + 2\delta$. So $H(c'') \leq 1 + 2\delta$.

Mate. For $|t| \leq \tau_5/\rho$, $p^*(f + t\rho g_{c''}) \leq s(\rho t) + \delta \rho^2 t^2 \leq s(t)$ as soon as $\delta \rho^2 \max(1, \tau_5/\rho) \leq (1 - \rho^2)/3$ (Lemma 3.6(a)). For $|t| \geq \tau_5/\rho$ we have $p^*(f + t\rho g_{c''}) \leq s(\rho t) + \rho|t|\delta' \leq s(t)$ as soon as $\rho\delta' \leq (1 - \rho^2)\min(1, \tau_5/\rho)/3$. So for δ small, $\rho g_{c''} \in \mathcal{C}(f)$ and $H(\rho c'') \leq \rho^2(1 + 2\delta) \leq 1$, i.e. $\rho g_{c''} \in \text{Cert}(f)$, at distance at most $\rho\delta'$ from ρg . Let $\delta \rightarrow 0$ and $\rho \rightarrow 1$. The last assertions follow from Corollary 3.9. $\square$

Corollary 4.11 (Split-contractive operators). *Let $S \in \mathcal{L}((c_0, q), \ell_2^2)$ and $\Phi \in \mathcal{L}(V, \ell_2^2)$ with $\|S\|_q \leq 1$, $\|\Phi\| \leq 1$, and suppose $T := S + \Phi \circ L$ has $\|T\|_p = 1$. Then $T \in \text{clNA}((c_0, p), \ell_2^2)$.*

Proof. After a rotation $T = (f, g)$ with $p^*(f) = 1$, $f = s_1 + L^*\varphi_1$, $g = s_2 + L^*\varphi_2$, where (s_1, s_2) is q -contractive and (φ_1, φ_2) is V -contractive. Then $q^*(s_1), \|\varphi_1\| \leq 1$, so by Proposition 1.8, $s_1 = a$ and $\varphi_1 = w$; the contractivity of the components gives $q^*(a + ts_2) \leq s(t)$ and $\|w + t\varphi_2\| \leq s(t)$ for all t . So g is locally split, and Theorem 4.10 applies. $\square$

Since $\text{NA} \cap \{S + \Phi \circ L\}$ is not dense [6, Section 3], the corollary shows that split operators are recovered by non-split norm-attaining ones.

4.4. Carriers with non-summable gaps.

Theorem 4.12 (Carrier mates). *Let $g \in \mathcal{C}(f)$, let $c^0 = (b^0, \omega^0)$ be a finite certificate, fix a block m and $c \in \mathbb{R} \setminus \{0\}$, and let $k_1, k_2, \dots$ be distinct strict non-peaks of w_m outside $\text{supp } \omega_m^0$, with $\lambda_i := \lambda_{k_i, m}$ and $\gamma_i := \text{gap}_m(k_i)$, such that*

- (a) $\gamma_{i+1}\lambda_{i+1} \leq \gamma_i\lambda_i/2$ for all i ;
- (b) $p^*(g - g_{c^0} - cu_{k_i, m}) \leq K\lambda_i$ for all i ;
- (c) $H_* := \max(h(b^0), \max_{m' \neq m} H_{m'}(\omega_{m'}^0), H_m(\omega_m^0) + c^2/(m^2 C_m)) \leq 1$;
- (d) $\sum_i \gamma_i = \infty$.

Then $g \in \text{clCert}(f)$. In particular this holds when $\inf_i \gamma_i > 0$ and $\|u_{k_i, m} - v\| \leq K'\lambda_i$ for some v with $g = g_{c^0} + cv$ (after passing to a subsequence for which (a) holds): cross mates with rates through carriers whose gaps are bounded below are recovered along every sequence.

Proof. Let $c_i := c^0 + (0, (c/\lambda_i)e_{k_i})$ in block m . Its d -coefficient in block m is $d_i = d_m^0 + c\Phi_m(k_i)w_m(k_i)/(mC_m) \rightarrow d_m^0$, and $\|D_m\omega_i\|^2 = \|D_m\omega_m^0\|^2 + c^2/m^2$ (because $\Phi_m(k_i)/\lambda_i = 1/m$), so $\limsup_i H(c_i) \leq H_* \leq 1$. The coordinatewise radius is $r_i = \min(r_*, \gamma_i\lambda_i/(2|c|))$ with $r_* > 0$ bounded below for large i , $\kappa(c_i)$ is bounded, and

$$e_i := p^*(g - g_{c_i}) \leq p^*(g - g_{c^0} - cu_{k_i, m}) + |d_i - d_m^0|p^*(R_m^*w_m) \leq K'\lambda_i.$$

Fix $\rho < 1$, choose η_0, κ_0, s_0 as in Theorem 4.1 with moreover $s_0 \sup_i \kappa(c_i) \leq \kappa_0$. For a stretch $i_0 \leq i \leq i_1$ put $\Gamma := \sum_{i_0}^{i_1} \gamma_i$ and $\mu_i := \gamma_i/\Gamma$. If $s \leq r_*/\rho$, then by (a)

$$\sum_{r_i < \rho s} \mu_i e_i \leq \frac{K'}{\Gamma} \sum_{\gamma_i \lambda_i < 2|c|\rho s} \gamma_i \lambda_i \leq \frac{4K'|c|\rho s}{\Gamma} \leq \frac{(1 - \rho^2)s}{12\rho}$$

once $\Gamma \geq 48K'|c|\rho^2/(1 - \rho^2)$, which is possible with i_0 arbitrarily large by (d). If $r_*/\rho < s \leq s_0$, then $\sum_i \mu_i e_i \leq K' \sup_{i \geq i_0} \lambda_i \leq (1 - \rho^2)r_*/(12\rho^2)$ for i_0 large (note $\lambda_i \rightarrow 0$), which gives (ii) as well; (iii) holds similarly, and (i) holds for i_0 large because $\limsup H(c_i) \leq 1$ and we may take $\eta_0 > 0$. Theorem 4.1 gives $\rho g_c \in \text{Cert}(f)$ within $\rho K' \sup_{i \geq i_0} \lambda_i$ of ρg . For the last assertion, $e_i \leq |c|K'\lambda_i + O(\lambda_i)$, and a subsequence with $\lambda_{i+1} \leq \lambda_i \inf \gamma/2$ satisfies (a). $\square$

Remark 4.13. If $\sum_i \gamma_i < \infty$ the argument breaks down: condition (ii) at $s \approx \gamma_i \lambda_i$ forces $\mu_i \lesssim \gamma_i$. Whether switching mates carried by such “super-near-threshold” coordinates, or by weak peaks (small margins), lie in $\text{clCert}(f)$ is open; their limit directions lie in the closed span of the non-peak vectors, so the linear obstruction of Section 6 does not apply to them.

5. SECOND-ORDER REBALANCING AND TAME SUPPORTS

In this section the block set I is finite. Let $f \in S_{p^*}$ have a normer $\eta \in S_{p^{**}}$; either $\eta = \xi$ is the bidual normer of an arbitrary f , or $\eta = x' \in S_p$ when $f = f'$ attains its norm. We use the forced data of Section 1 at η , writing $c_\eta := q^{**}(\eta)$ (so $c_\eta = q_0$ when $\eta = \xi$), $\hat{z} := \eta/c_\eta = z + Ue$, $\zeta_m := R_m^{**}\eta$ and $\sigma_m := |\zeta_m|_m$, so that $c_\eta + \sum_m \sigma_m = 1$.

5.1. Slices, excess and tame points.

Lemma 5.1 (Bidual slice). *Let $g \in \mathcal{C}(f)$. Then $g(\eta) = 0$, and for every $\chi \in X^{**}$ with $\chi(f) = 0$,*

$$\chi(g)^2 \leq \varphi(\chi)(2 + \varphi(\chi)), \quad \varphi(\chi) := p^{**}(\eta + \chi) - 1.$$

In particular, if $\varphi(s\chi) = o(s^2)$ as $s \rightarrow 0^+$, then $\chi(g) = 0$.

Proof. For $\theta \in X^{**}$ and $t \in \mathbb{R}$, $\theta(f) + t\theta(g) = \theta(f + tg) \leq p^{**}(\theta)s(t)$; the supremum of the left side divided by $s(t)$ over t is $\sqrt{\theta(f)^2 + \theta(g)^2}$, so $\theta(f)^2 + \theta(g)^2 \leq p^{**}(\theta)^2$. With $\theta = \eta$ we get $g(\eta) = 0$, and with $\theta = \eta + \chi$ we get $1 + \chi(g)^2 \leq (1 + \varphi(\chi))^2$. For the last claim apply this to $s\chi$ and divide by s^2 . $\square$

Lemma 5.2 (Excess splitting). *For $\chi \in X^{**}$ put $\Delta(\eta + \chi) := p^{**}(\eta + \chi) - f(\eta + \chi) \geq 0$. Then*

$$\Delta(\eta + \chi) = E_q(\chi) + \sum_m E_m(R_m^{**}\chi),$$

$$E_q(\chi) := q^{**}(\eta + \chi) - c_\eta - a(\chi), \quad E_m(\delta) := |\zeta_m + \delta|_m - \sigma_m - w_m(\delta),$$

and all these terms are nonnegative. If $f(\chi) = 0$, then $\Delta(\eta + \chi) = \varphi(\chi)$.

Proof. $p^{**}(\eta) = 1 = f(\eta) = c_\eta + \sum_m \sigma_m$, $f = a + \sum_m R_m^* w_m$ and $p^{**} = q^{**} + \sum_m |R_m^{**} \cdot|_m$; nonnegativity holds because a and w_m have norm one. $\square$

Lemma 5.3 (Flatness on free coordinates). *If $\chi \in \ell_\infty$ is supported in J and $|\chi_j| \leq c_\eta(1 - |z_j|)$ for all j , then $E_q(\chi) = 0$ and $a(\chi) = 0$.*

Proof. $\eta + \chi = c_\eta((z + \chi/c_\eta) + Ue)$ with $\|z + \chi/c_\eta\|_\infty \leq 1$, so $q^{**}(\eta + \chi) \leq c_\eta$; and $q^{**}(\eta + \chi) \geq a(\eta + \chi) = c_\eta$ because $J \cap F = \emptyset$. $\square$

Lemma 5.4 (Overshoot bound). *Fix a block and drop m . Let $\delta = \mu\zeta + \delta'$ with $\mu > -1$, $\delta' \in \ell_1$ supported in P and $\sum_{k \in P} s_k \delta'_k = 0$, where $s_k := \text{sign } w(k)$. Then*

$$E(\delta) \leq 2 \sum_{k \in P} (-s_k \delta'_k - (1 + \mu)|\zeta| |\alpha(k)|)_+ \leq 2 \sum_{k \in P} (|\delta'_k| - (1 + \mu)\lambda_k \mu_k)_+.$$

Proof. By Lemma 1.6, $\zeta + \delta = [(1 + \mu)|\zeta|\alpha + \delta'] + (1 + \mu)|\zeta|D(Dw/C)$. Using $|x + y| = |x| + sy + 2(-sy - |x|)_+$ for $s \in \{\pm 1\}$ with $sx = |x|$, the ℓ_1 -norm of the bracket is $(1 + \mu)|\zeta| + \sum_P s_k \delta'_k + 2\text{ov} = (1 + \mu)|\zeta| + 2\text{ov}$, where ov is the sum in the statement; the second term lies in $(1 + \mu)|\zeta|D(B_{\ell_2})$. Hence $|\zeta + \delta| \leq (1 + \mu)|\zeta| + 2\text{ov}$, while $w(\delta) = \mu|\zeta| + M \sum_P s_k \delta'_k = \mu|\zeta|$. The second inequality is (2). $\square$

Note that $|(R_m \chi)(k)| = \lambda_{k,m} |u_{k,m}(\chi)| \leq \lambda_{k,m} q(\chi)$ for $\chi \in X$.

Definition 5.5 (Tame points, certificate span). Let $P_m^\circ := \{k \in P_m : \mu_{k,m} > 0\}$ and $Dg_m := P_m \setminus P_m^\circ$ (degenerate peaks), $\bar{Q}_m := Q_m \cup Dg_m$. The normer η satisfies *margin sparsity* (MS) if $\sum_{k \in P_m^\circ, \mu_{k,m} < s} \Phi_m(k) = o(s)$ as $s \rightarrow 0^+$ for every m . It is *C-tame* if $a \in c_{00}$, K is finite, every $\bar{Q}_m$ is finite, and (MS) holds. (At $\eta = x' \in c_0$ the first two conditions are automatic.) The *certificate span* at f is

$$S(f) := \left\{ b + \sum_m R_m^*(\omega_m - d_m(\omega_m)w_m) : b \in \ell_1(F), b(\eta) = 0, \omega_m \in c_{00}(Q_m) \right\},$$

$$d_m(\omega) := \frac{\langle D_m w_m, D_m \omega \rangle}{C_m},$$

and its elements are called *balanced finite certificates* (the base part need not satisfy $\|b/a\|_\infty < \infty$ when F is infinite). Equivalently $S(f) = (\ell_1(F) \cap \eta^\perp) + \text{span}\{y_{k,m} : k \in Q_m, m \in I\}$ with $y_{k,m} := \lambda_{k,m}u_{k,m} - (\Phi_m(k)^2 w_m(k)/C_m)R_m^* w_m$. For $g \in S(f)$ with data (b, ω) put $H_b := h(b)$, $H_m := H_m(\omega_m)$ and

$$\Gamma_w(g) := c_\eta H_b + \sum_m \sigma_m H_m, \quad \Gamma_{\max}(g) := \max(H_b, \max_m H_m).$$

When F and all Q_m are finite the data (b, ω) of $g \in S(f)$ are unique (Lemma 1.7), so Γ_w is well defined; in general we regard Γ_w as a function of the data. Since $c_\eta + \sum_m \sigma_m = 1$, $\Gamma_w \leq \Gamma_{\max}$.

Theorem 5.6 (Fibres at C-tame points). *If the normer η of f is C-tame, then $\mathcal{C}(f) \subseteq S(f)$; that is, every mate is a balanced finite certificate $g = b + \sum_m R_m^*(\omega_m - d_m w_m)$ with $\text{supp } b \subseteq F$, $b(\eta) = 0$ and ω_m supported in the finite set Q_m . In particular $\mathcal{C}(f)$ is finite dimensional.*

Proof. J is cofinite. By Lemma 1.7(b) with $P'_m := P_m^\circ$ (cofinite, hence infinite) and $\pi_m := \sum_{k \in P_m} s_k \lambda_{k,m} u_{k,m}$ (so that $\pi_m(\chi) = \sum_{P_m} s_k (R_m \chi)_k$; the degenerate peaks are among the finitely many $u_{k,m}$, $k \in \bar{Q}_m$), the restrictions to $c_{00}(J)$ of the finite family

$$\Psi := \{u_{k,m} : k \in \bar{Q}_m, m \in I\} \cup \{\pi_m : m \in I\}$$

are linearly independent. Note $R_m^* w_m = M_m \pi_m + \sum_{k \in Q_m} w_m(k) \lambda_{k,m} u_{k,m}$. We use repeatedly: if $\chi \in X$ and $R_m \chi = \mu \zeta_m + \delta'$ where δ' vanishes on $\bar{Q}_m$ and has zero peak sum, then by Lemma 5.4 and (MS), $E_m(t\mu \zeta_m + t\delta') \leq 2m \sum_{k \in P_m^\circ} \Phi_m(k) (|t|B - \frac{1}{2}\mu_{k,m})_+ = o(t^2)$ for $|t\mu| \leq \frac{1}{2}$, where B bounds $|\delta'_k|/\lambda_{k,m}$.

Step 1: $\mathcal{C}(f) \subseteq W := \text{span}\{e_j^* : j \in F \cup K\} + \text{span } \Psi$. Let $\nu \in c_{00}(J)$ with $\psi(\nu) = 0$ for all $\psi \in \Psi$. Then $a(\nu) = 0$, $R_m \nu$ vanishes on $\bar{Q}_m$, and its peak sum is $\pi_m(\nu) = 0$; also $f(\nu) = a(\nu) + \sum_m w_m(R_m \nu) = 0$. For small $|t|$, $E_q(t\nu) = 0$ (Lemma 5.3: ν is finitely supported in J) and $E_m(tR_m \nu) = o(t^2)$, so $\varphi(t\nu) = o(t^2)$ and $g(\nu) = 0$ for every $g \in \mathcal{C}(f)$ (Lemma 5.1). By linear algebra, the restriction of g to $c_{00}(J)$ is a combination of the restrictions of Ψ ; hence g minus that combination is supported in $F \cup K$.

Step 2: unique form. Since π_m and the $u_{k,m}$ are images under R_m^* of $s1_{P_m}$ and $\lambda_{k,m}^{-1} e_k$, every $g \in W$ can be written $g = b + \sum_m R_m^*(\omega_m + c_m w_m)$ with $\text{supp } b \subseteq F \cup K$ and $\omega_m \in c_{00}(\bar{Q}_m)$; the representation is unique by Lemma 1.7(a) and because $w_m \neq 0$ on the infinite set P_m° .

Step 3: no contact components. Let $j \in K$. Choose $\nu_J \in c_{00}(J)$ with $\psi(\nu_J) = z_j \psi(e_j)$ for $\psi \in \Psi$ and put $\nu := -z_j e_j + \nu_J$, so $\psi(\nu) = 0$ for $\psi \in \Psi$. For small $s > 0$, $\eta + s\nu = c_\eta((z - sz_j e_j/c_\eta + s\nu_J/c_\eta) + Ue)$ has cube part of norm ≤ 1 (the

contact moves inward), so $q^{**}(\eta + s\nu) \leq c_\eta = a(\eta + s\nu)$ and $E_q(s\nu) = 0$. As in Step 1, $E_m(sR_m\nu) = o(s^2)$ and $f(\nu) = 0$. Hence $g(\nu) = 0$. But $g(\nu) = -z_j b_j$, because $b(\nu_J) = 0$ and $(\omega_m + c_m w_m)(R_m\nu) = 0$. So $b_j = 0$.

Step 4: no degenerate components. Let $k_0 \in Dg_{m_0}$ and $s_0 := \text{sign } w_{m_0}(k_0)$. Choose $\nu \in c_{00}(J)$ with $u_{k_0, m_0}(\nu) = s_0$ and $\psi(\nu) = 0$ for all other $\psi \in \Psi$. Then $R_{m_0}\nu$ vanishes on $\bar{Q}_{m_0} \setminus \{k_0\}$, equals $s_0 \lambda_{k_0, m_0}$ at k_0 , and has zero peak sum. For $s > 0$ the first bound of Lemma 5.4 contributes nothing at k_0 (there $-s_0 \cdot s \cdot s_0 \lambda_{k_0, m_0} < 0$) and nothing at the other degenerate peaks; the non-degenerate peaks give $o(s^2)$ by (MS). Other blocks and the base are as in Step 1. So $\varphi(s\nu) = o(s^2)$ for $s > 0$ and $g(\nu) = 0$; but $g(\nu) = \omega_{m_0}(k_0) \lambda_{k_0, m_0} s_0$. So $\omega_{m_0}(k_0) = 0$.

Step 5: balance. Fix m_0 . Choose $\nu_J \in c_{00}(J)$ with $\psi(\nu_J) = -\psi(\eta)/\sigma_{m_0}$ for $\psi \in \{u_{k, m_0} : k \in \bar{Q}_{m_0}\} \cup \{\pi_{m_0}\}$ and $\psi(\nu_J) = 0$ for the other elements of Ψ , and put $\nu := \eta + \nu_J \in X^{**}$. Then $f(\nu) = 1 + \sum_m w_m(R_m\nu_J) = 1 - w_{m_0}(\zeta_{m_0})/\sigma_{m_0} = 0$. The base: $\eta + t\nu = (1+t)(\eta + t\nu_J/(1+t))$, so $E_q(t\nu) = 0$ for small $|t|$ by Lemma 5.3 and homogeneity. Blocks $m \neq m_0$: $R_m^{**}(t\nu) = t\zeta_m + tR_m\nu_J$ with $R_m\nu_J$ vanishing on $\bar{Q}_m$ and of zero peak sum. Block m_0 : $R_{m_0}^{**}(t\nu) = t(1 - 1/\sigma_{m_0})\zeta_{m_0} + t\delta''$ with $\delta'' := R_{m_0}\nu_J + \zeta_{m_0}/\sigma_{m_0}$ vanishing on $\bar{Q}_{m_0}$ and of zero peak sum. Hence $\varphi(t\nu) = o(t^2)$, and $g(\nu) = 0$ by Lemma 5.1. Writing $v_m := \omega_m + c_m w_m$ we compute $g(\nu) = g(\eta) + b(\nu_J) + \sum_m v_m(R_m\nu_J) = -v_{m_0}(\zeta_{m_0})/\sigma_{m_0}$, since $v_m(\delta) = 0$ whenever δ vanishes on $\bar{Q}_m$ and has zero peak sum. Thus $v_m(\zeta_m) = 0$ for all m . As ω_m lives on Q_m , where $\zeta_m(k) = \sigma_m \Phi_m(k)^2 w_m(k)/C_m$, we have $\omega_m(\zeta_m) = \sigma_m d_m(\omega_m)$, and $v_m(\zeta_m) = \sigma_m(d_m + c_m)$; hence $c_m = -d_m$. Finally $b(\eta) = g(\eta) - \sum_m v_m(\zeta_m) = 0$. $\square$

Remark 5.7. At a norm-attaining f' with C-tame normer x' (i.e. finitely many strict non-peaks and degenerate peaks, and (MS)), Theorem 5.6 says $\mathcal{C}(f') \subseteq S(f')$. This is a *linear obstruction at the approximants*: if $f_n \rightarrow f$ are norm attaining with C-tame normers, then $\text{Li}_n \mathcal{C}(f_n) \subseteq \text{Li}_n S(f_n)$. It will be used in Section 6.

5.2. Transfer peaks and the mass-weighted coefficient. In this subsection $\eta = \xi$. The *canonical truncations* of f are defined when $a \in c_{00}$: let $z_j^N := z_j$ for $j \in F \cup [1, N]$ and $z_j^N := 0$ otherwise, $\tilde{x}_N := z^N + Ue \in c_0$, $x'_N := \tilde{x}_N/p(\tilde{x}_N)$ and $f'_N := \nabla p(x'_N)$. By Proposition 1.9(c), $q(\tilde{x}_N) = 1$, $\nabla q(\tilde{x}_N) = a$ and $f'_N = a + \sum_m R_m^* w'_{m, N}$ with $w'_{m, N} = J_m(R_m x'_N)$; by Proposition 1.11, $f'_N \rightarrow f$. We write $c_N := q(x'_N) \rightarrow q_0$, and $M'_{m, N}, C'_{m, N}, \sigma'_{m, N}, \dots$ for the data at x'_N ; they converge to those at ξ (Proposition 1.10).

Theorem 5.8 (Transfer peaks). *Let $f \in S_{p^*}$ with $a \in c_{00}$, and let $g = b + \sum_m R_m^*(\omega_m - d_m w_m) \in S(f)$ be a balanced finite certificate such that (f, g) is contractive and $\Gamma_w(g) \leq 1$. Then for every $\rho \in (0, 1)$ there are $g'_N \in \mathcal{C}(f'_N)$, N large, with $g'_N \rightarrow \rho g$. In particular $(f, \rho g) \in \text{clNA}((c_0, p), \ell_2^2)$, and all such mates are recovered simultaneously along the canonical truncations, which do not depend on g .*

Proof. Fix $\rho \in (0, 1)$ and $\eta \in (0, \frac{1}{2}]$ with $\rho^2(\Gamma_w(g) + 4\eta) \leq \rho$, and put $L := \Gamma_w(g)/2 + \eta$. Let $\gamma_0 := \frac{1}{2} \min_m \min_{k \in \text{supp } \omega_m} \text{gap}_m(k) > 0$.

Step 1: the transported natural certificate. Put $b'_N := b - (b(x'_N)/c_N)a$, $d'_{m, N} := \langle D_m w'_{m, N}, D_m \omega_m \rangle / C'_{m, N}$, $v'_{m, N} := \omega_m - d'_{m, N} w'_{m, N}$ and $g'_N := b'_N + \sum_m R_m^* v'_{m, N}$. Since $b(x'_N) \rightarrow b(\xi) = 0$, $d'_{m, N} \rightarrow d_m$ and $R_m^* w'_{m, N} \rightarrow R_m^* w_m$, we have $g'_N \rightarrow g$. For N large every $k \in \text{supp } \omega_m$ has $M'_{m, N} - |w'_{m, N}(k)| \geq \gamma_0$. *Base:* b'_N is supported

in the finite set F , where $z^N = \text{sign } a$. The computation in the proof of Lemma 3.3, with $\hat{z}$ replaced by $\tilde{x}_N = z^N + Ue$, gives $q^*(a + tb'_N) = 1 + t b'_N(\tilde{x}_N) + \nu \Psi(tU^*b'_N/\nu)$ as long as no coordinate of $a + tb'_N$ changes sign on F ; and $b'_N(\tilde{x}_N) = b'_N(x'_N)/c_N = 0$. Since $P^\perp U^*b'_N = P^\perp U^*b$ and $b'_N \rightarrow b$, the bounds for Ψ give $q^*(a + tb'_N) \leq 1 + \frac{t^2}{2} H_b \theta(t)$, where here and below $\theta(t) \rightarrow 1$ as $t \rightarrow 0$, uniformly in large N . *Blocks:* Lemma 3.4(c),(d) at x'_N give $\|w'_{m,N} + tv'_{m,N}\|_\infty \leq (1 - td'_{m,N})M'_{m,N}$ and $\|D_m(w'_{m,N} + tv'_{m,N})\| \leq C'_{m,N} + td'_{m,N}M'_{m,N} + \frac{t^2}{2} H'_{m,N}C'_{m,N}\theta(t)$, with $H'_{m,N} := (\|D_m\omega_m\|^2 - d'^2_{m,N})/C'_{m,N} \rightarrow H_m$.

Step 2: transfer peaks at ξ . Fix m and put $\varsigma_m := \text{sign}(H_m/2 - L) \in \{-1, 0, 1\}$; if $\varsigma_m = 0$ set $\tau_m := 0$ and skip this step. Choose $\gamma_m \in (\max_{k \in \text{supp } \omega_m} |w_m(k)|, M_m)$ with $|w_m(k)| \neq \gamma_m$ for all k , let $\Lambda_m := \{k \notin \text{supp } \omega_m : |w_m(k)| \geq \gamma_m\} \supseteq P_m$, $y_m^0 := \text{sign}(w_m)1_{\Lambda_m}$, $\hat{\pi}_m := R_m^* y_m^0$ and $\tau_{*,m} := (\hat{\pi}_m(\xi)/q_0)a - \hat{\pi}_m$, so $\tau_{*,m}(\xi) = 0$. For $\delta_0 \in (0, 1]$ let $T_m := \varsigma_m \tau_{*,m} + \delta_0(q^*(\tau_{*,m}) + 1)a$ and $\ell_m := q^*(T_m)$; then $T_m(\xi) = \delta_0(q^*(\tau_{*,m}) + 1)q_0 > 0$. For $\delta_1 \in (0, T_m(\xi)/(2\ell_m q_0))$ choose, by density of the tails of $(u_{k,m})_k$, an index $k_* = k_{*,m} \notin \text{supp } \omega_m$ with $q^*(u_{k_*,m} - T_m/\ell_m) \leq \delta_1$, so large that $\ell_m \Phi_m(k_*)M_m/(mC_m) \leq \frac{1}{2}$ and $\vartheta_m \Phi_m(k_*) < T_m(\xi)/(2\ell_m)$. Then $u_{k_*,m}(\xi) \geq T_m(\xi)/\ell_m - \delta_1 q_0 \geq T_m(\xi)/(2\ell_m) > \vartheta_m \Phi_m(k_*)$, so k_* is a non-degenerate peak with $w_m(k_*) = M_m$ and margin $\mu_* > 0$. Put

$$y_m := y_m^0 + \varsigma_m \frac{\ell_m}{\lambda_{k_*,m}} e_{k_*}, \quad R_m^* y_m = \hat{\pi}_m + \varsigma_m \ell_m u_{k_*,m},$$

$$e_{\infty,m} := R_m^* y_m - \frac{(R_m^* y_m)(\xi)}{q_0} a.$$

Since $R_m^* y_m = (\text{multiple of } a) + \varsigma_m(\ell_m u_{k_*,m} - T_m)$, we get $e_{\infty,m} = \varsigma_m[(\ell_m u_{k_*,m} - T_m) - \frac{(\ell_m u_{k_*,m} - T_m)(\xi)}{q_0} a]$ and $q^*(e_{\infty,m}) \leq 2\ell_m \delta_1$. Also $\ell_m \mu_* \leq \ell_m u_{k_*,m}(\xi) \leq T_m(\xi) + \ell_m \delta_1 q_0$. Hence the *inefficiency* $\epsilon_{1,m} := \ell_m \mu_*/q_0 + q^*(e_{\infty,m}) \leq \delta_0(q^*(\tau_{*,m}) + 1) + 3\ell_m \delta_1$ can be made as small as we wish by choosing first δ_0 and then δ_1 (note $\ell_m \leq (1 + \delta_0)(q^*(\tau_{*,m}) + 1)$). Put $e_{y,m} := 1 + \langle D_m y_m, D_m w_m \rangle / C_m$. As $y_m^0 w_m \geq 0$ and $\ell_m \Phi_m(k_*)M_m/(mC_m) \leq \frac{1}{2}$, we have $e_{y,m} \geq \frac{1}{2}$. Finally let $\tau_m := (H_m/2 - L)/e_{y,m}$, whose sign is ς_m , and note $|\tau_m| \leq H_m + 2$ independently of δ_0, δ_1, k_* . We choose δ_0, δ_1 so that $\sum_m (H_m + 2)\epsilon_{1,m} \leq \eta/2$.

Step 3: the limiting levels. Put $\Lambda_b := H_b/2 + \sum_m (\tau_m \sigma_m e_{y,m}/q_0 + |\tau_m| \epsilon_{1,m})$ and $\Lambda_m := H_m/2 - \tau_m e_{y,m} = L$. Using $\sum_m \sigma_m = 1 - q_0$,

$$q_0 \Lambda_b = \frac{\Gamma_w}{2} - L(1 - q_0) + q_0 \sum_m |\tau_m| \epsilon_{1,m}, \quad \text{so} \quad \Lambda_b = L - \frac{\eta}{q_0} + \sum_m |\tau_m| \epsilon_{1,m} \leq L.$$

Step 4: transport of the transfer data. For N large let $\Lambda_{m,N} := \{k \notin \text{supp } \omega_m : |w'_{m,N}(k)| \geq \gamma_m\}$ and $y_{m,N} := \text{sign}(w'_{m,N})1_{\Lambda_{m,N}} + \varsigma_m(\ell_m/\lambda_{k_*,m})e_{k_*}$. For large N , k_* is a peak of $w'_{m,N}$ of sign $+1$ with margin $\mu'_{*,N} \rightarrow \mu_*$, and the peak set of $w'_{m,N}$ is contained in $\Lambda_{m,N}$. Since $w'_{m,N} \rightarrow w_m$ coordinatewise and $|w_m(k)| \neq \gamma_m$, $y_{m,N} \rightarrow y_m$ coordinatewise and boundedly; hence $R_m^* y_{m,N} \rightarrow R_m^* y_m$ in ℓ_1 and $D_m y_{m,N} \rightarrow D_m y_m$ in ℓ_2 . Put $\beta_{m,N} := ((R_m^* y_{m,N})(x'_N)/c_N)a$, $e_{m,N} := R_m^* y_{m,N} - \beta_{m,N} \rightarrow e_{\infty,m}$, and $e_{y,m,N} := 1 + \langle D_m y_{m,N}, D_m w'_{m,N} \rangle / C'_{m,N} \rightarrow e_{y,m}$. Writing $R_m x'_N = \sigma'_{m,N}(\alpha' + D_m^2 w'_{m,N}/C'_{m,N})$ (Lemma 1.6) and using (2) at k_* , one obtains the bookkeeping identity

$$(6) \quad y_{m,N}(R_m x'_N) = \sigma'_{m,N} e_{y,m,N} + \varsigma_m \ell_m \mu'_{*,N}.$$

Indeed $\langle \text{sign}(w')1_{\Lambda_{m,N}}, \alpha' \rangle = 1$, and

$$\frac{\ell_m}{\lambda_{k_*}}(R_m x'_N)(k_*) = \ell_m u_{k_*}(x'_N) = \ell_m \mu'_{*,N} + \frac{\ell_m}{\lambda_{k_*}} \frac{\sigma' \Phi(k_*)^2 M'}{C'},$$

the last term being the contribution of e_{k_*} to $\sigma' \langle Dy_{m,N}, Dw' \rangle / C'$.

Step 5: the decomposition. For $t \in \mathbb{R}$ put $\varepsilon_m := \tau_m \rho^2 t^2$ and

$$A(t) := a + t \rho b'_N + \sum_m \varepsilon_m R_m^* y_{m,N}, \quad W_m(t) := w'_{m,N} + t \rho v'_{m,N} - \varepsilon_m y_{m,N},$$

so that $A(t) + \sum_m R_m^* W_m(t) = f'_N + t \rho g'_N$. *Base.* By (6), $A(t) = (1 + E)a + t \rho b'_N + \sum_m \varepsilon_m e_{m,N}$ with $E := \rho^2 t^2 \sum_m (\tau_m \sigma'_{m,N} e_{y,m,N} + |\tau_m| \ell_m \mu'_{*,N}) / c_N$. By homogeneity and Step 1,

$$q^*(A(t)) \leq 1 + E + \frac{\rho^2 t^2}{2(1 + E)} H_b \theta(t) + \rho^2 t^2 \sum_m |\tau_m| q^*(e_{m,N}) = 1 + \rho^2 t^2 (\Lambda_b + o(1)),$$

where $o(1) \rightarrow 0$ as $t \rightarrow 0$ and $N \rightarrow \infty$. *Blocks, sup norm.* For $|t|$ small and N large, $\|W_m(t)\|_\infty \leq (1 - t \rho d'_{m,N}) M'_{m,N} - \varepsilon_m$: on $\Lambda_{m,N} \setminus \{k_*\}$ the coordinates are $(1 - t \rho d') w'(k) - \varepsilon_m \text{sign } w'(k)$, whose moduli are at most this bound; at k_* the value is $(1 - t \rho d') M' - \varepsilon_m (1 + \varsigma_m \ell_m / \lambda_{k_*})$, which lies between $\pm((1 - t \rho d') M' - \varepsilon_m)$ once $|\varepsilon_m| \ell_m / \lambda_{k_*} \leq M'$ (here $\varsigma_m \varepsilon_m \geq 0$ is used); on $\text{supp } \omega_m$ the moduli are at most $(1 - t \rho d')(M' - \gamma_0) + \rho |t| \|\omega_m\|_\infty$; elsewhere at most $(1 - t \rho d') \gamma_m$; both are below the bound for small $|t|$ and large N because $M'_{m,N} \rightarrow M_m > \gamma_m$. *Blocks, Hilbert part.* With $h_0 := D_m(w' + t \rho v')$, inequality (5) gives $\|h_0 - \varepsilon_m D_m y_{m,N}\| \leq \|h_0\| - \varepsilon_m \langle h_0 / \|h_0\|, D_m y_{m,N} \rangle + O(t^4)$, and $\langle h_0 / \|h_0\|, D_m y_{m,N} \rangle = \langle D_m w', D_m y_{m,N} \rangle / C' + O(|t|)$ uniformly in N . Together with Step 1,

$$N_m(W_m(t)) \leq 1 + \rho^2 t^2 \left(\frac{H'_{m,N} \theta(t)}{2} - \tau_m e_{y,m,N} \right) + O(|t|^3) = 1 + \rho^2 t^2 (L + o(1)).$$

By Lemma 1.5 and Step 3, there are $t_4 > 0$ and N_4 with $p^*(f'_N + t \rho g'_N) \leq 1 + \rho^2 t^2 (L + \eta) \leq 1 + \frac{t^2}{2} \rho^2 (\Gamma_w + 4\eta) \leq 1 + \rho t^2 / 2 \leq s(t)$ for $|t| \leq t_5 := \min(t_4, 2\sqrt{1 - \rho}, 1)$ and $N \geq N_4$.

Step 6: large t . Since (f, g) is contractive, $p^*(f'_N + t \rho g'_N) \leq s(\rho t) + \varepsilon_N + \rho |t| \eta_N$ with $\varepsilon_N := p^*(f'_N - f) \rightarrow 0$ and $\eta_N := p^*(g'_N - g) \rightarrow 0$; by Lemma 3.6(a) this is at most $s(t)$ for $|t| \geq t_5$ and N large. Hence $\rho g'_N \in \mathcal{C}(f'_N)$ and $\rho g'_N \rightarrow \rho g$. $\square$

Remark 5.9. (a) If $\Gamma_{\max}(g) \leq 1$, no transfer peaks are needed ($\tau_m = 0$); this is the natural-certificate case, and together with Corollary 3.10 it re-proves the local-certificate theorem of [8]. (b) The transfer peak is the genuinely infinite-dimensional ingredient: lowering a block's sup norm requires lowering every near-peak coordinate, and the image $\hat{\pi}_m$ of such a uniform lowering is a fixed vector of Y which the base cannot absorb; density of the tails of $(u_{k,m})_k$ provides a robust deep peak whose functional converts it into a multiple of a with arbitrarily small inefficiency. In finite-dimensional models of the construction this mechanism is absent and the second-order coefficient can be strictly larger than Γ_w . (c) The same estimates performed at ξ instead of x'_N show that $\limsup_{t \rightarrow 0} (p^*(f + tg)^2 - 1)/t^2 \leq \Gamma_w(g)$ for every balanced finite certificate when $a \in c_{00}$; we do not use this.

Remark 5.10 (Γ_w and shifts). For every shifted certificate $c = (b, \omega, \theta)$ at f one has $H^{\text{sh}}(c) \geq \Gamma_w(g_c)$. Indeed, if $\Lambda := H^{\text{sh}}(c)$, then $2\theta_m \leq \Lambda - H_m$ for every m , and $\Lambda \geq h(b) - 2 \sum_m \theta_m \sigma_m / q_0 \geq h(b) - \sum_m (\Lambda - H_m) \sigma_m / q_0$; multiplying by q_0 and

using $q_0 + \sum_m \sigma_m = 1$ gives $\Lambda \geq \Gamma_w$. Thus Theorem 5.8 covers every balanced shifted certificate whose direction lies in $S(f)$.

5.3. The sharp second-order coefficient at tame supports.

Lemma 5.11 (Exact local block formula). *Fix a block and drop m ; recall $\sum_k \Phi(k)^2 < 1$. Let $y \in \ell_1$, let $P \neq \emptyset$ be a set of indices with complement Q , and $s \in \{\pm 1\}^P$. Put $S := \sum_{k \in P} s_k y_k$, $Y := (\sum_{k \in Q} y_k^2 / \Phi(k)^2)^{1/2}$ and $F_2 := \sum_{k \in P} \Phi(k)^2 \in (0, 1)$, and assume $0 < S$, $Y < \infty$ and $Y \leq S$. Define*

$$\Lambda(S, Y) := \frac{S - \sqrt{F_2} \sqrt{S^2 - (1 - F_2)Y^2}}{1 - F_2}, \quad \lambda := \Lambda(S, Y), \quad \rho_0 := (S/\lambda - 1)/F_2.$$

Then $\lambda > 0$ and

$$(7) \quad |y| \leq \lambda + 2 \sum_{k \in P} (\lambda \rho_0 \Phi(k)^2 - s_k y_k)_+.$$

Moreover, if $\zeta \neq 0$ with norming functional w , and (P, s) are its peak set and signs, then with $c_Q := \|D(w1_Q)\|_2^2$: $S_P(\zeta) = |\zeta|(1 + F_2 M/C)$, $Y_Q(\zeta) = |\zeta|\sqrt{c_Q}/C < S_P(\zeta)$, $\Lambda(S_P(\zeta), Y_Q(\zeta)) = |\zeta|$, $\lambda \rho_0 = |\zeta|M/C$, and at this point

$$\partial_S \Lambda = M, \quad \partial_Y \Lambda = CY/|\zeta|, \quad \partial_S^2 \Lambda = \frac{\sqrt{F_2} Y^2}{(S^2 - (1 - F_2)Y^2)^{3/2}} = \frac{C c_Q}{|\zeta| F_2}.$$

Proof. $\Delta := S^2 - (1 - F_2)Y^2 \geq F_2 S^2 > 0$; λ is the smaller root of $(1 - F_2)l^2 - 2Sl + (S^2 + F_2 Y^2) = 0$, i.e. of $(S - l)^2 + F_2 Y^2 = F_2 l^2$, and $\lambda > 0$ because $\sqrt{F_2 \Delta} < S$. The root equation reads $\rho_0^2 F_2 + Y^2/\lambda^2 = 1$. Define $h \in \ell_2$ by $h_k := \rho_0 \Phi(k) s_k$ ($k \in P$), $h_k := y_k/(\lambda \Phi(k))$ ($k \in Q$); then $\|h\|_2 = 1$, and $\alpha := y/\lambda - Dh$ vanishes on Q , with $\|\alpha\|_1 = \sum_P s_k \alpha_k + 2 \sum_P (-s_k \alpha_k)_+ = S/\lambda - \rho_0 F_2 + \frac{2}{\lambda} \sum_P (\lambda \rho_0 \Phi(k)^2 - s_k y_k)_+$, and $S/\lambda - \rho_0 F_2 = 1$. Hence $y = \lambda \alpha + \lambda Dh \in \lambda \|\alpha\|_1 (B_{\ell_1} + D(B_{\ell_2}))$, which is (7).

At ζ , Lemma 1.6 gives $s_k \zeta_k = |\zeta|(|\alpha_k| + \Phi(k)^2 M/C)$ on P and $\zeta_k = |\zeta|\Phi(k)^2 w_k/C$ on Q , whence the formulas for S and Y ; $Y < S$ because $\sqrt{c_Q} \leq C$ and $F_2 M > 0$. Using $C^2 = M^2 F_2 + c_Q$ one checks that $l = |\zeta|$ satisfies $(S - l)^2 + F_2 Y^2 = F_2 l^2$; since $|\zeta|(1 - F_2) < S$, it is the smaller root. Then $\rho_0 = M/C$, and from the root equation $\sqrt{F_2 \Delta} = S - (1 - F_2)\lambda = \lambda F_2(1 + \rho_0)$, i.e. $\sqrt{\Delta} = |\zeta|\sqrt{F_2}/C$. Differentiating, $\partial_S \Lambda = (1 - \sqrt{F_2} S/\sqrt{\Delta})/(1 - F_2) = (1 - C - F_2 M)/(1 - F_2) = M$, $\partial_Y \Lambda = \sqrt{F_2} Y/\sqrt{\Delta} = CY/|\zeta|$, and $\partial_S^2 \Lambda = \sqrt{F_2}(S^2 - \Delta)/((1 - F_2)\Delta^{3/2}) = \sqrt{F_2} Y^2/\Delta^{3/2} = C c_Q/(|\zeta| F_2)$. $\square$

Proposition 5.12 (Lower bound). *Let I be finite and let $f \in S_{p^*}$ satisfy: $a \in c_{00}$, K is finite, every Q_m is finite, there are no degenerate peaks, and (MS) holds at ξ . Then for every balanced finite certificate $g \in S(f)$,*

$$\liminf_{t \rightarrow 0} \frac{p^*(f + tg) - 1}{t^2} \geq \frac{\Gamma_w(g)}{2}.$$

In particular $\Gamma_w(g) \leq 1$ for every $g \in \mathcal{C}(f) \cap S(f)$.

Proof. Write $g = b + \sum_m R_m^* v_m$, $v_m = \omega_m - d_m w_m$. For $\chi, \chi_2 \in X$ put $\eta_t := \xi + t\chi + t^2 \chi_2$. Since $g(\xi) = 0$ and $p^{**}(\eta_t) = f(\eta_t) + \Delta(\eta_t)$ with $f(\eta_t) = 1 + O(t)$,

$$(8) \quad p^*(f + tg) \geq \frac{(f + tg)(\eta_t)}{p^{**}(\eta_t)} = 1 + t^2 g(\chi) - \Delta(\eta_t) + O(t^3) \quad \text{whenever } \Delta(\eta_t) = O(t^2).$$

By Lemma 5.2, $\Delta(\eta_t) = E_q(t\chi + t^2 \chi_2) + \sum_m E_m(t\delta_{1,m} + t^2 \delta_{2,m})$ with $\delta_{i,m} := R_m \chi_i$.

Base test. Let $E_F := \text{span}\{U^*e_j^* : j \in F\} \ni e$ and fix $k_\perp \in E_F$ with $\langle k_\perp, e \rangle = 0$. Put $\kappa_2 := \nu\|k_\perp\|^2/(2q_0)$, $\kappa'_2 := -\kappa_2\|a\|_1/\nu$, and let $k_{2\perp} \in E_F$ solve $\langle k_{2\perp}, U^*e_j^* \rangle = -\kappa_2 z_j - \kappa'_2(Ue)_j$ for $j \in F$ (the Gram matrix of the independent vectors $U^*e_j^*$ is invertible); then $\langle k_{2\perp}, e \rangle = (-\kappa_2\|a\|_1 - \kappa'_2\nu)/\nu = 0$. Put $k_2 := \kappa'_2 e + k_{2\perp}$, so that $(Uk_2)_j = -\kappa_2 z_j$ on F . Let $\chi := Uk_\perp + \chi_c$ with $\chi_c \in c_{00}(J)$ to be chosen, and $\chi_2 := (Uk_2)1_{\mathbb{N} \setminus F}$. With $\mathbf{h} := q_0 e + tk_\perp + t^2 k_2$ and $Z := \eta_t - U\mathbf{h} = q_0 z + t\chi_c - t^2(Uk_2)1_F$ we have $|Z_j| = q_0 + t^2\kappa_2$ on F and $|Z_j| \leq q_0$ off F for small t (χ_c moves finitely many free coordinates), while $\|\mathbf{h}\| = q_0 + t^2(\kappa'_2 + \|k_\perp\|^2/(2q_0)) + O(t^3) = q_0 + t^2\kappa_2(1 - \|a\|_1)/\nu + O(t^3) = q_0 + t^2\kappa_2 + O(t^3)$, using $\|a\|_1 + \nu = 1$. Since $B_{q^{**}} = B_{\ell_\infty} + U(B_H)$, $q^{**}(\eta_t) \leq q_0 + t^2\kappa_2 + O(t^3)$; and $a(\chi) = \langle U^*a, k_\perp \rangle = 0 = a(\chi_2)$. Hence $E_q(t\chi + t^2\chi_2) \leq t^2\nu\|k_\perp\|^2/(2q_0) + O(t^3)$, and $b(\chi) = \langle U^*b, k_\perp \rangle$.

Block test. Fix m and drop it. Apply (7) to $y := \zeta + t\delta_1 + t^2\delta_2$ with the peak set and signs of ζ . As Q is finite, $G(y) := \Lambda(S_P(y), Y_Q(y))$ is a smooth function of the finitely many numbers $(S_P(y), (y_k)_{k \in Q})$ near those of ζ (Λ depends smoothly on (S, Y^2)). By Lemma 5.11 its differential at ζ is $\delta \mapsto MS_P(\delta) + \sum_{k \in Q} w_k \delta_k = w(\delta)$, so $G(y) = |\zeta| + w(t\delta_1 + t^2\delta_2) + \frac{t^2}{2}\mathcal{H}(\delta_1) + O(t^3)$, where $\mathcal{H}$ is the Hessian of G at ζ . The overshoot: $\lambda(y)\rho_0(y) = |\zeta|M/C + O(t)$ and, by (2), $s_k y_k = \lambda_k \mu_k + |\zeta|\Phi(k)^2 M/C + ts_k \delta_{1,k} + t^2 s_k \delta_{2,k}$ on P , with $|\delta_{i,k}| \leq \lambda_k q(\chi_i)$; so each overshoot term is at most $\lambda_k(|tB - \mu_k|_+)$ for a constant B , and their sum is $o(t^2)$ by (MS), all peaks being non-degenerate. Hence $E_m(t\delta_1 + t^2\delta_2) \leq \frac{t^2}{2}\mathcal{H}_m(\delta_{1,m}) + o(t^2)$.

Decoupling. $\mathcal{H}_m(\delta)$ and $v_m(\delta)$ depend only on $\sigma_1 := S_{P_m}(\delta)$ and $\delta|_{Q_m}$. Since $F \cup K$ is finite, Lemma 1.7(b) (with $P'_m = P_m$, which is cofinite) allows us to choose $\chi_c \in c_{00}(J)$ so that, for the fixed $k_\perp$, the finitely many numbers $u_{k,m}(\chi)$ ($k \in Q_m$) and $\pi_m(\chi) = S_{P_m}(R_m\chi)$ take arbitrary prescribed values. By (8),

$$\liminf_{t \rightarrow 0} \frac{p^*(f + tg) - 1}{t^2} \geq \frac{1}{2} \left[\sup_{k_\perp} \left(2\langle U^*b, k_\perp \rangle - \frac{\nu}{q_0} \|k_\perp\|^2 \right) + \sum_m \sup_\delta (2v_m(\delta) - \mathcal{H}_m(\delta)) \right].$$

Base supremum. $U^*b \in E_F$; the maximizer is $k_\perp = (q_0/\nu)P^\perp U^*b \in E_F \cap e^\perp$, with value $(q_0/\nu)\|P^\perp U^*b\|^2 = q_0 H_b$.

Block supremum. Drop m . Suppose first $c_Q > 0$, so $Y > 0$. Write $\zeta' := D^{-1}\delta_Q$, $\hat{h} := D^{-1}\zeta_Q/Y = Dw_Q/\sqrt{c_Q}$, $\sigma_2 := \langle \hat{h}, \zeta' \rangle$, $\zeta'_\perp := \zeta' - \sigma_2 \hat{h}$ and $r := Y\sigma_1 - S\sigma_2$. Since Λ is 1-homogeneous, its Hessian is $(\partial_S^2 \Lambda/Y^2)(Y, -S)^{\otimes 2}$, and $Y_Q(\zeta + \delta) = Y + \sigma_2 + \|\zeta'_\perp\|^2/(2Y) + O(\|\delta\|^3)$; with Lemma 5.11, $\mathcal{H}(\delta) = (C/|\zeta|)\|\zeta'_\perp\|^2 + (\partial_S^2 \Lambda/Y^2)r^2$. Next, with $\tilde{\omega} := (\omega - dw)1_Q$, $v(\delta) = \langle D\tilde{\omega}, \zeta'_\perp \rangle + (\langle D\tilde{\omega}, \hat{h} \rangle \sigma_2 - dM\sigma_1)$. The linear form in brackets vanishes at $(\sigma_1, \sigma_2) = (S, Y)$, since there it equals $v(\zeta) = \omega(\zeta) - d|\zeta| = 0$ ($\omega(\zeta) = |\zeta|d$ because $\zeta = |\zeta|D^2w/C$ on Q); hence it equals $-(dM/Y)r$. As $\zeta'_\perp \in \hat{h}^\perp$ and r are independent variables,

$$\begin{aligned} \sup_\delta (2v - \mathcal{H}) &= \frac{|\zeta|}{C} \|P_{\hat{h}^\perp} D\tilde{\omega}\|^2 + \frac{d^2 M^2}{\partial_S^2 \Lambda} = \frac{|\zeta|}{C} \left(\|D\omega\|^2 - \frac{d^2 C^2}{c_Q} \right) + \frac{d^2 M^2 |\zeta| F_2}{C c_Q} \\ &= \frac{|\zeta|}{C} (\|D\omega\|^2 - d^2), \end{aligned}$$

where we used $\langle D\omega, D\omega \rangle = dC$, $\|P_{\hat{h}^\perp} D\tilde{\omega}\|^2 = \|D\omega\|^2 - d^2 C^2/c_Q$ (a direct computation) and $C^2 = c_Q + M^2 F_2$. This is $\sigma_m H_m$. If $c_Q = 0$, then $w_Q = 0$, $d = 0$, $\Lambda(S, \cdot)$ depends on Y^2 with $\partial \Lambda / \partial (Y^2) = C/(2|\zeta|)$ at ζ , $\mathcal{H}(\delta) = (C/|\zeta|)\|D^{-1}\delta_Q\|^2$, $v(\delta) = \langle D\omega, D^{-1}\delta_Q \rangle$, and the supremum is $(|\zeta|/C)\|D\omega\|^2 = \sigma_m H_m$ directly.

Summing, $\liminf_{t \rightarrow 0} (p^*(f + tg) - 1)/t^2 \geq \frac{1}{2}(q_0 H_b + \sum_m \sigma_m H_m) = \Gamma_w(g)/2$. If $g \in \mathcal{C}(f)$, then $(p^*(f + tg) - 1)/t^2 \leq (s(t) - 1)/t^2 \rightarrow \frac{1}{2}$. $\square$

Theorem 5.13 (Tame supports are recoverable). *Let I be finite and let $f \in S_{p^*}$ have a C -tame normer without degenerate peaks, i.e. $a \in c_{00}$, K finite, every Q_m finite, $Dg_m = \emptyset$ and (MS). Then $\mathcal{C}(f) \subseteq \text{Li}_N \mathcal{C}(f'_N)$ for the canonical truncations f'_N of f . In particular $f \in \mathcal{R}$.*

Proof. By Theorem 5.6, every $g \in \mathcal{C}(f)$ is a balanced finite certificate; by Proposition 5.12, $\Gamma_w(g) \leq 1$; by Theorem 5.8, $\rho g \in \text{Li}_N \mathcal{C}(f'_N)$ for every $\rho < 1$, and the lower limit is closed. Corollary 2.3 gives $f \in \mathcal{R}$. $\square$

Remark 5.14 (On the hypotheses). (a) The finiteness of K enters Theorem 5.13 twice: in Step 1 of Theorem 5.6 and in the decoupling step of Proposition 5.12, both of which need the set J of free coordinates to be cofinite so that Lemma 1.7(b) applies. Lemma B does not exclude elements of $\text{span}\{u_{k,m}, \pi_m\} \subseteq Y$ supported in $F \cup K$ when K is infinite, and then the lower bound may fail: in a finite-dimensional analogue in which every off-support coordinate is a contact, the exact second-order coefficient of a pure block certificate is at most 0.296 on both sides while $\Gamma_w = 0.485$ (re-splitting through the contacts is free at first order for one sign of t). Whether a finite certificate with $\Gamma_2 \leq 1 < \Gamma_w$ exists at a support with infinitely many contacts in Martín's space is *open*; Theorem 5.8 would not recover it. (b) Earlier formulations assumed $K = \emptyset$; Step 3 of Theorem 5.6 removes this. (c) The existence of tame non-attaining supports for a *given* admissible T is not known in general (they form a meagre set in the bidual face, by the residuality of half-peaks in [6, Section 3]). For the operator T of Section 6 the canonical truncations of the special first row are C -tame norm-attaining points (Proposition 6.8).

6. THE RESONANCE OBSTRUCTION

Throughout this section I is finite.

6.1. The linear obstruction. For $f \in S_{p^*}$ let $\mathfrak{B}(f) := \{g \in \mathcal{C}(f) \cap S(f) : \Gamma_w(g) \leq 1\}$ if $a \in c_{00}$ (then the data of $g \in S(f)$ are unique by Lemma 1.7(a)), and $\mathfrak{B}(f) := \emptyset$ otherwise. The *intrinsic class* $\mathcal{K}(f)$ is the closed convex hull of $\text{Cert}^{\text{sh}}(f) \cup \mathfrak{B}(f)$, and the *defect* is $\text{Def}(f) := \mathcal{C}(f) \setminus \mathcal{K}(f)$.

Proposition 6.1 (Intrinsic classes). (a) $\mathcal{K}(f)$ contains every mate shown to be recoverable in Sections 3–5: $\text{cl Cert}(f)$ (hence the mates of Corollaries 3.10 and 4.2 and Theorems 4.6, 4.10, 4.12), $\text{cl Cert}^{\text{sh}}(f)$ and $\mathfrak{B}(f)$. (b) $\mathcal{K}(f) \subseteq \text{cl } S(f)$, a closed linear subspace of $\{g : g(\xi) = 0\}$. (c) If $a \in c_{00}$, then $\mathcal{K}(f) \subseteq \text{Li}_N \mathcal{C}(f'_N)$ for the canonical truncations; in any case $\text{cl Cert}^{\text{sh}}(f) \subseteq \text{Li}_n \mathcal{C}(f_n)$ for every sequence $f_n \rightarrow f$.

Proof. (a) is a summary of the quoted results ($\text{Cert}(f) \subseteq \text{Cert}^{\text{sh}}(f)$). (b) Conditions (C1), (C2) say exactly that $g_c \in S(f)$, for shifted certificates as well, and $\mathfrak{B}(f) \subseteq S(f)$ by definition. (c) Theorem 3.13 for Cert^{sh} , Theorem 5.8 for $\mathfrak{B}(f)$, and closedness and convexity of lower limits of convex sets. $\square$

Corollary 6.2 (Linear obstruction). $\mathcal{C}(f) \setminus \text{cl } S(f) \subseteq \text{Def}(f)$. If F and every Q_m are finite, then $S(f)$ is finite dimensional, and a mate g lies outside $\mathcal{K}(f)$ as soon as $g \notin \text{span}\{e_j^* : j \in F\} + \text{span}\{u_{k,m} : k \in Q_m\} + \text{span}\{R_m^* w_m\}$.

When some block has infinitely many strict non-peaks whose vectors are spread in $\ker \xi$, the closure of $S(f)$ may be all of $\ker \xi$ and the obstruction is void. It bites at first rows with few non-peaks. We now construct such a first row with an infinite-dimensional fibre.

6.2. An admissible operator with an exact resonance. Let $\nu_j := \|U^*e_j^*\| \rightarrow 0$ and $J_0 := \{j \geq 2 : \nu_j \leq \frac{1}{2}\}$ (cofinite). Put $\alpha := 1/(1 + \nu_1)$, $a := \alpha e_1^*$ (so $q^*(a) = 1$), $e := U^*e_1^*/\nu_1$, and note $(Ue)_1 = \nu_1$. Fix a bijection $l : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ with $l(1, 1) = 1$ and $l(n, m) < l(n', m)$ for $n < n'$, pairwise disjoint infinite sets $S_l \subseteq J_0$ ($l \in \mathbb{N}$), and put $h_l := \sum_{s \in S_l} 2^{-s} e_s^*$, $c_l := 2^{-(l-1)^2}$ (so $c_1 = 1$ and $\sum_{l' \geq l} c_{l'} \leq 2c_l$). Let $l_0 := l(2, 1)$, $K' := S_{l_0}$,

$$z := e_1 + 1_{K'} \in \ell_\infty, \quad \hat{z} := z + Ue, \quad z^{(n)} := e_1 + 1_{K' \cap [1, n]}, \quad x^{(n)} := z^{(n)} + Ue \in c_0.$$

Note $\hat{z}_1 = x_1^{(n)} = 1 + \nu_1 = 1/\alpha$. For $k \geq 2$, $(k, m) \neq (2, 1)$, put $\pi_{k, m} := 2^{1-m-k} c_{l(k, m)}/c_{l(1, m)} \leq \frac{1}{4}$ and $\rho_{l(k, m)} := \sqrt{\pi_{k, m}}$, and call (k, m) *exceptional* if $\pi_{k, m} \geq \epsilon_*^2$, $\epsilon_* := 1/(72(1 + \|U\|))$ (finitely many). Let $\delta_{l(1, m)} := \min(2^{-l}, 1/(8(1 + \|U\|)))$, $\delta_{l_0} := 1$, $\delta_l := \min(2^{-l}, 1/(16(1 + \|U\|)))$ for exceptional l , and $\delta_l := \min(2^{-l}, \rho_l/(1 + \|U\|))$ otherwise. Fix a dense sequence $(y^{(i)})_{i \geq 1}$ in $c_{00} \cap S_{q^*}$ with $y^{(1)} := e_1^*/q^*(e_1^*)$, let $\Lambda(i) := \{n : \text{supp } y^{(i)} \cap K' \cap (n, \infty) \neq \emptyset\}$ (finite), and call i *allowed* at a non-exceptional $l = l(k, m)$ ($k \geq 2$, $l \neq l_0$) if

$$\text{supp } y^{(i)} \cap S_l = \emptyset, \quad 2c_l \leq 2^{-2s} c_{l'} \delta_{l'} \text{ for } s \in \text{supp } y^{(i)} \cap S_{l'}, \quad l' \neq l, \quad 3\rho_l(2|\Lambda(i)| + 3) \leq \frac{1}{8}.$$

Each i is allowed at all large l , and $i = 1$ is allowed everywhere ($\Lambda(1) = \emptyset$ and $9\rho_l \leq \frac{1}{8}$). Fix a sequence (i_n) in which every positive integer occurs infinitely often, and let $i(n, m) := i_n$ if i_n is allowed at $l(n, m)$ and $i(n, m) := 1$ otherwise. Define, with n_l the q^* -norm of the bracket,

- $u_{1, m} := (e_1^*/q^*(e_1^*) + \delta_l h_l)/n_l$ and, for exceptional (k, m) , $u_{k, m} := (e_1^*/q^*(e_1^*) + \delta_l h_l)/n_l$ ($l = l(k, m)$);
- $u_{2, 1} := (-\kappa e_1^* + h_{l_0})/n_{l_0}$, with $\kappa := (\|h_{l_0}\|_1 + \langle U^* h_{l_0}, e \rangle)/(1 + \nu_1)$;
- for the remaining (k, m) , with $l = l(k, m)$ and $y := y^{(i(k, m))}$: $u_{k, m} := (y + s_l \alpha e_1^* + \delta_l h_l)/n_l$, where $|s_l| \leq 3\rho_l(2|\Lambda(i)| + 3)$ is chosen so that $|y(x) + s_l| \geq 3\rho_l$ for every x in the finite set $\{\hat{z}\} \cup \{x^{(n)} : n \in \Lambda(i)\}$;

and $Te_{n, m} := c_{l(n, m)} u_{n, m}$. The choice of s_l is possible: the excluded values form $|\Lambda(i)| + 1$ intervals of length $6\rho_l$ inside an interval of length $6\rho_l(2|\Lambda(i)| + 3)$; and $(\alpha e_1^*)(x) = \alpha x_1 = 1$ for all these x .

Lemma 6.3 (The operator T). *T is admissible, $\Phi_m(k) = 2^{-m-k} c_{l(k, m)}$, and:*

- (P1) $\text{supp } u_{2, 1} = \{1\} \cup K'$, $u_{2, 1}(1) < 0 < u_{2, 1}(j)$ for $j \in K'$, and $u_{2, 1}(\hat{z}) = 0$;
- (P2) $|u_{1, m}(x)| \geq 7/9$ for $x \in X_0 := \{\hat{z}\} \cup \{x^{(n)} : n \in \mathbb{N}\}$ and every m ;
- (P3) $|u_{k, m}(x)| \geq \frac{3}{2} \rho_{l(k, m)}$ for $x \in X_0$, every non-exceptional (k, m) with $k \geq 2$, $(k, m) \neq (2, 1)$, and $|u_{k, m}(x)| \geq \frac{1}{2}$ for exceptional (k, m) ; in both cases $|u_{k, m}(x)| > \pi_{k, m}$.

Proof. Norms. $q^*(h_l) \leq (1 + \|U\|)\|h_l\|_1 \leq 1 + \|U\|$, so $q^*(\delta_l h_l) \leq \frac{1}{8}$ for $l = l(1, m)$, $\leq \frac{1}{16}$ for exceptional l , and $\leq \rho_l \leq \frac{1}{72}$ otherwise; also $q^*(s_l \alpha e_1^*) = |s_l| \leq \frac{1}{8}$. Hence $n_l \in [\frac{7}{8}, \frac{9}{8}]$ in the first two cases and $n_l \in [\frac{3}{4}, \frac{5}{4}]$ in the last. (T-a): $q^*(Te_{n, m}) = c_{l(n, m)} \leq 1 = c_1$.

(P1): $|\langle U^* h_{l_0}, e \rangle| \leq \sum_{s \in K'} 2^{-s} \nu_s \leq \|h_{l_0}\|_1/2$, so $\kappa > 0$, and $(-\kappa e_1^* + h_{l_0})(\hat{z}) = -\kappa(1 + \nu_1) + \|h_{l_0}\|_1 + \langle U^* h_{l_0}, e \rangle = 0$ because $z = 1$ on K' .

(P2), (P3): every $x \in X_0$ satisfies $x_1 = 1/\alpha$, $x = Ue$ on $\bigcup_{l \neq l_0} S_l$ and $|x_s| \leq \|U\|$ there. Hence $(e_1^*/q^*(e_1^*))(x) = 1$ and $|\delta_l h_l(x)| \leq \delta_l \|U\|$; this gives $|u_{1,m}(x)| \geq (1 - \frac{1}{8})/\frac{9}{8} = \frac{7}{9}$ and, for exceptional (k, m) , $\geq (1 - \frac{1}{16})/\frac{17}{16} > \frac{1}{2} \geq 2\pi_{k,m}$. For the remaining (k, m) and $n \notin \Lambda(i)$, $x^{(n)} - \hat{z} = -1_{K' \cap (n, \infty)}$ is annihilated by y , by e_1^* and by h_l , so the choice of s_l gives $|(y + s_l \alpha e_1^*)(x)| \geq 3\rho_l$ for all $x \in X_0$, and $|u_{k,m}(x)| \geq (3\rho_l - \rho_l)/\frac{5}{4} = \frac{8}{5}\rho_l > \rho_l \geq \pi_{k,m}$.

(T-d): fix m and i . Infinitely many n have $i(n, m) = i$, and along them $\|u_{n,m} - y^{(i)}\|_1 \rightarrow 0$, because $\rho_l, \delta_l \rightarrow 0$, $|s_l| \leq 3\rho_l(2|\Lambda(i)| + 3) \rightarrow 0$ and $n_l \rightarrow 1$ ($l = l(n, m)$). Hence every $y^{(i)}$ is a limit point of $(u_{n,m})_n$.

(T-b), (T-c): let $x \in \ell_1(\mathbb{N} \times \mathbb{N})$, indexed by l , with $\mathcal{Z} := \sum_l x_l c_l u_l \in c_{00}$, and suppose $x_{l'} \neq 0$. Write $u_l = (y_l + \delta_l h_l)/n_l$ with $y_l \in c_{00}$; the y_l of the first two items and the multiples of e_1^* live on $\{1\}$, and $1 \notin \bigcup S_l$. For $s \in S_{l'}$ only $h_{l'}$ among the h_l is nonzero at s , and $y_{l'}(s) = 0$ (allowedness). If $s \in \text{supp } y_l$ for some l , let $L(s)$ be the least such l ; allowedness at $L(s)$ gives $2c_{L(s)} \leq 2^{-2s} c_{l'} \delta_{l'}$, and with $|y_l(s)| \leq 1$, $n_l \geq \frac{3}{4}$,

$$|\mathcal{Z}(s)| \geq \frac{|x_{l'}| c_{l'} \delta_{l'} 2^{-s}}{n_{l'}} - \frac{4}{3} \|x\|_\infty \sum_{l \geq L(s)} c_l \geq c_{l'} \delta_{l'} 2^{-s} \left(\frac{|x_{l'}|}{n_{l'}} - \frac{4}{3} \|x\|_\infty 2^{-s} \right) > 0$$

for all large $s \in S_{l'}$. As $S_{l'}$ is infinite, $\mathcal{Z} \notin c_{00}$, a contradiction. Hence $x = 0$: T is injective and $Y \cap c_{00} = \{0\}$. $\square$

The construction is flexible (any finite set of prescribed vectors can be inserted, and the targets may be the same in all blocks, as in Martín's proof); only (T-a)–(T-d) and (P1)–(P3) are used below.

Lemma 6.4 (The first row). *With T as in Lemma 6.3 and any finite $I \ni 1$, put $\xi := \hat{z}/p^{**}(\hat{z})$ and $f := a + L^* J_V(L^{**}\xi)$. Then $f \in S_{p^*}$ is not norm attaining, ξ is its normer, $q_0 = 1/p^{**}(\hat{z})$, its forced data are $a = \alpha e_1^*$, $w = J_V(L^{**}\xi)$ and $z = e_1 + 1_{K'}$, $F = \{1\}$, and $K = K'$ is infinite. Moreover $(2, 1)$ is a strict non-peak with $w_1(2) = 0$, every other coordinate (k, m) , $m \in I$, is a non-degenerate peak, and $S(f) = \mathbb{R}u_{2,1}$.*

Proof. $\hat{z} \in B_{\ell_\infty} + U(B_H) = B_{q^{**}}$ and $a(\hat{z}) = \alpha \hat{z}_1 = 1 = q^*(a)$, so $q^{**}(\hat{z}) = 1$ and $p^{**}(\hat{z}) = 1 + \|L^{**}\hat{z}\|$. Then $f(\xi) = (1 + \|L^{**}\hat{z}\|)/p^{**}(\hat{z}) = 1$ and $p^*(f) \leq \max(q^*(a), \|J_V\|) = 1$ by Lemma 1.5. Proposition 1.8 gives the forced data, and $\xi \notin c_0$ shows that f does not attain its norm (a norming point in X would be the unique normer).

Blocks: $\zeta_m(k) = \lambda_{k,m} q_0 u_{k,m}(\hat{z})$, and $\sigma_m \leq \|\zeta_m\|_1 \leq q_0 m 2^{-m}$. By Lemma 1.6, off P_m we have $|\zeta_m(k)| = \sigma_m \Phi_m(k)^2 |w_m(k)|/C_m \leq \sigma_m \Phi_m(k)$, i.e. $|u_{k,m}(\xi)| \leq \sigma_m/m$, and $|u_{k,m}(\xi)| > \vartheta_m \Phi_m(k)$ implies $k \in P_m$ with positive margin. By (P2), $|u_{1,m}(\xi)| \geq \frac{7}{9} q_0 > q_0 2^{-m} \geq \sigma_m/m$, so $1 \in P_m$; hence $C_m \geq \Phi_m(1) M_m$ and $\vartheta_m \Phi_m(k) \leq \sigma_m \Phi_m(k)/(m \Phi_m(1)) \leq q_0 \pi_{k,m}$. By (P3), every $(k, m) \neq (2, 1)$ with $k \geq 2$ is a peak with positive margin; and $u_{1,m}(\xi)$ exceeds $q_0/2 \geq \vartheta_m \Phi_m(1)$. Finally $u_{2,1}(\xi) = 0$ by (P1), so $\zeta_1(2) = 0$, $2 \notin P_1$ and $w_1(2) = 0$. Thus $Q_1 = \{2\}$, $Q_m = \emptyset$ for $m \neq 1$, and $S(f) = (\ell_1(\{1\}) \cap \xi^\perp) + \mathbb{R}u_{2,1} = \mathbb{R}\lambda_{2,1} u_{2,1}$ because $\xi_1 \neq 0$ and $w_1(2) = 0$. $\square$

Proposition 6.5 (Two-piece mates). *Let $u := u_{2,1}$, $\lambda_0 := \lambda_{2,1} = \Phi_1(2)$, $c > 0$, $v := cu$, and for $K_1 \subseteq K'$ put $K_2 := K' \setminus K_1$, $\beta := (v1_{K_1})(\hat{z})$ and $g_{K_1} := v1_{K_1} - \beta a$.*

There is $c_* > 0$, depending only on $\|U\|, \nu_1, M_1, C_1, \lambda_0$, such that for $0 < c \leq c_*$ every g_{K_1} lies in $\mathcal{C}(f)$, with the admissible decompositions

$$f + tg = (a + tg) + L^*w \quad (t \geq 0), \quad f + tg = (a + t(g - v)) + L^*(w + tD) \quad (t \leq 0),$$

where $D := (c/\lambda_0)e_2$ in block 1 (so $L^*D = v$).

Proof. We use $\|xe + y\| \leq x + \langle e, y \rangle + \|y\|^2/(2x)$ for $x > 0, y \in H$, which follows from $(\langle e, y \rangle + \|y\|^2/(2x))^2 \geq 0$. Let $|t| \leq 1$ and $c \leq c_1 := \min(\alpha/4, 1/(4(1 + \|U\|)))$; then $|\beta| \leq \|v\|_1 \|\hat{z}\|_\infty \leq c(1 + \|U\|) \leq \frac{1}{4}$. Put $\nu := \|U^*a\| = \alpha\nu_1$.

Side $t \geq 0$. $a + tg = (1 - t\beta)\alpha e_1^* + tv1_{K_1}$, with $1 - t\beta \geq \frac{3}{4}$ and $v > 0$ on K_1 . So $q^*(a + tg) \leq (1 - t\beta)(\alpha + \nu) + t(\|v1_{K_1}\|_1 + \langle e, h_1 \rangle) + \frac{2}{3}t^2\|h_1\|^2/\nu$ with $h_1 := U^*(v1_{K_1})$. Here $\alpha + \nu = 1$ and $\|v1_{K_1}\|_1 + \langle e, h_1 \rangle = (v1_{K_1})(z + Ue) = \beta$ since $z = 1$ on K_1 . Hence $q^*(a + tg) \leq 1 + \frac{2}{3}t^2\|h_1\|^2/\nu$, and the block part is w .

Side $t = -s \leq 0$. $g - v = -\beta a - v_1e_1^* - v1_{K_2}$ with $v_1 := v(1) < 0, |v_1| \leq c$, so $a + t(g - v) = (\alpha(1 + s\beta) + sv_1)e_1^* + sv1_{K_2}$ with first coefficient $\geq \alpha/2$. As before, $q^*(a + t(g - v)) \leq (1 + s\beta) + sv_1/\alpha + s(v1_{K_2})(\hat{z}) + s^2\|h_2\|^2/\nu$ with $h_2 := U^*(v1_{K_2})$, and the first-order bracket equals $sv(\hat{z}) = scu(\hat{z}) = 0$ (P1). In block 1, $W := w_1 + tD$ differs from w_1 only at 2, where $|W(2)| = sc/\lambda_0 \leq M_1$ if $c \leq M_1\lambda_0$; the peaks keep the value M_1 , so $\|W\|_\infty = M_1$, and $\|D_1W\|^2 = C_1^2 + s^2c^2$ because $w_1(2) = 0$ and $\Phi_1(2)/\lambda_0 = 1$. Thus $N_1(W) \leq 1 + s^2c^2/(2C_1)$.

Conclusion. For $|t| \leq 1, p^*(f + tg) \leq 1 + t^2 \max(\|U\|^2c^2/\nu, c^2/(2C_1)) \leq 1 + \frac{3}{8}t^2 \leq s(t)$ if moreover $c^2\|U\|^2 \leq 3\nu/8$ and $c^2 \leq \frac{3}{4}C_1$. For $|t| \geq 1, p^*(f + tg) \leq 1 + |t|q^*(g) \leq 1 + |t| \cdot 2c(1 + \|U\|) \leq 1 + |t|(\sqrt{2} - 1) \leq s(t)$ if $c \leq 0.2/(1 + \|U\|)$, since $(s(t) - 1)/|t|$ increases. Take c_* the minimum of these bounds. $\square$

Theorem 6.6 (Nonempty defect). *For the admissible operator T of Lemma 6.3 and every finite $I \ni 1$, the first row f of Lemma 6.4 satisfies $\mathcal{K}(f) \subseteq \mathbb{R}u_{2,1}$, while $\mathcal{C}(f) \ni g_{K_1}$ for every $K_1 \subseteq K'$. If $\emptyset \neq K_1 \neq K'$, then $g_{K_1} \notin \mathbb{R}u_{2,1}$, so $g_{K_1} \in \text{Def}(f)$. The mates $g_{\{j\}} = cu_{2,1}(j)(e_j^* - \hat{z}_j\alpha e_1^*), j \in K'$, are finitely supported, and the defect spans an infinite-dimensional space.*

Proof. By Lemma 6.4 and Proposition 6.1(b), $\mathcal{K}(f) \subseteq \mathbb{R}u_{2,1}$. If $g_{K_1} = \mu u_{2,1}$, comparing coordinates $j \in K'$ (where a vanishes and $u_{2,1}(j) > 0$) gives $c1_{K_1}(j) = \mu$ for all $j \in K'$, which forces $K_1 \in \{\emptyset, K'\}$. $\square$

Remark 6.7. (a) The mechanism is an *exact resonance*: the block vector $u_{2,1}$ of a strict non-peak with $w = 0$ is, as a base functional, supported on $F \cup K'$ with the contact signs on K' , and $u_{2,1}(\hat{z}) = 0$; hence the block carrier and the contact vector represent the same functional, and splitting the contacts between the two signs of t produces switching mates. For $K_1 \in \{\emptyset, K'\}$, g_{K_1} is a two-sided certificate. (b) The same estimates show that every g supported in $\{1\} \cup K'$ with $g(\xi) = 0$ and $\sup_{j \in K'} |g_j/u_{2,1}(j)|$ small is a mate, so $\mathcal{C}(f)$ contains an infinite-dimensional slab; conversely one can show that every mate is supported in $\{1\} \cup K'$ with $g_j/u_{2,1}(j)$ bounded on K' (we do not use this). (c) Since K' is infinite, f is not C-tame, and Theorem 5.6 does not apply at f ; the infinite contact set is unavoidable (Proposition 6.9).

6.3. Non-recovery along canonical truncations. The defect of Theorem 6.6 is defined relative to the known intrinsic classes; a priori its elements could still be recovered along every sequence. For the canonical truncations this is not the case.

Proposition 6.8 (Non-recovery). *Let T , I and f be as in Theorem 6.6, and let $f_n := \nabla p(x^{(n)})$, $x^{(n)} = z^{(n)} + Ue$, be the canonical truncations of f . Then:*

- (a) $f_n \in \text{NA} \cap S_{p^*}$ and $f_n \rightarrow f$;
- (b) for large n the normer $x'_n := x^{(n)}/p(x^{(n)})$ of f_n is C -tame, with $Q_{n,1} = \{2\}$, $Q_{n,m} = \emptyset$ for $m \neq 1$, and no degenerate peaks;
- (c) $\mathcal{C}(f_n) \subseteq \mathbb{R}y^{(n)}$, where $y^{(n)} := \lambda_0 u_{2,1} - (\Phi_1(2)^2 w_1^{(n)}(2)/C_1^{(n)}) R_1^* w_1^{(n)} \rightarrow \lambda_0 u_{2,1}$;
- (d) $\text{Li}_n \mathcal{C}(f_n) \subseteq \mathbb{R}u_{2,1}$. In particular, for $\emptyset \neq K_1 \neq K'$ and $\rho \in (0, 1]$, $\rho g_{K_1} \notin \text{Li}_n \mathcal{C}(f_n)$: the mates g_{K_1} are not recovered along the canonical truncations.

Proof. (a) $z^{(n)} \in c_{00}$, $\|z^{(n)}\|_\infty \leq 1$ and $z_1^{(n)} = 1 = \text{sign } a_1$, so $\nabla q(x^{(n)}) = a$ and $f_n \in \text{NA} \cap S_{p^*}$ by Proposition 1.9(c); $z^{(n)} \rightarrow z$ coordinatewise, and Proposition 1.11 gives $f_n \rightarrow f$.

(b) Let $c'_n := q(x'_n) = 1/p(x^{(n)}) \rightarrow q_0 > 0$. Since $|u_{k,m}(x^{(n)})| \leq q(x^{(n)}) = 1$, $\sigma'_{n,m} \leq c'_n m 2^{-m}$. Exactly as in the proof of Lemma 6.4, using (P2) and (P3) at $x^{(n)} \in X_0$: 1 is a peak of every block, $\vartheta'_{n,m} \Phi_m(k) \leq c'_n \pi_{k,m}$, and every $(k, m) \neq (2, 1)$ is a peak with margin $\mu'_{k,m} \geq c'_n \rho_{l(k,m)}$ if (k, m) is non-exceptional with $k \geq 2$, and $\mu'_{k,m} \geq c'_n/4$ otherwise. As $\rho_{l(k,m)} = (2\Phi_m(k)/c_{l(1,m)})^{1/2} \geq (2\Phi_m(k))^{1/2}$ and $\Phi_m(k+1) \leq \Phi_m(k)/2$, for $s < c'_n/4$ we get $\sum_{\mu'_{k,m} < s} \Phi_m(k) \leq \sum_{\Phi_m(k) < s^2/(2c_n'^2)} \Phi_m(k) \leq s^2/c_n'^2 = o(s)$: (MS) holds. Next, $u_{2,1}(x^{(n)}) = u_{2,1}(\hat{z}) - \sum_{j \in K', j > n} u_{2,1}(j) \rightarrow 0$, while $\vartheta'_{n,1} \Phi_1(2) \rightarrow \vartheta_1 \Phi_1(2) > 0$ (Proposition 1.10); so for large n , $|u_{2,1}(x'_n)| < \vartheta'_{n,1} \Phi_1(2)$ and $(2, 1)$ is a strict non-peak. Finally $a \in c_{00}$ and the contact set of x'_n is $K' \cap [1, n]$, which is finite.

(c) By Theorem 5.6 at $\eta = x'_n$, $\mathcal{C}(f_n) \subseteq S(f_n) = (\ell_1(\{1\}) \cap (x'_n)^\perp) + \mathbb{R}y^{(n)} = \mathbb{R}y^{(n)}$, since $x'_n(1) \neq 0$. By Proposition 1.10, $w_1^{(n)}(2) \rightarrow w_1(2) = 0$ and $C_1^{(n)} \rightarrow C_1 > 0$, so $y^{(n)} \rightarrow \lambda_0 u_{2,1}$.

(d) If $\gamma_n y^{(n)} \rightarrow g$, then (γ_n) is bounded because $\|y^{(n)}\| \rightarrow \lambda_0 \|u_{2,1}\| > 0$; along a subsequence $\gamma_n \rightarrow \gamma$ and $g = \gamma \lambda_0 u_{2,1}$. The last claim follows from Theorem 6.6. $\square$

Thus, for this T , the mates recovered along *every* sequence are contained in the certificate line $\mathbb{R}u_{2,1}$: universal recovery fails, and a recovery of the switching mates must use norm-attaining approximants whose certificate spans grow, for instance by putting base mass on the contact windows (Section 7). By Remark 5.7, along any sequence of norm-attaining approximants with C -tame normers, recovery is a linear-algebra design problem: $\text{Li}_n \mathcal{C}(f_n) \subseteq \text{Li}_n S(f_n)$.

6.4. Exact resonances.

Proposition 6.9 (Exact resonances need infinite one-sided sets). *Let $f \in S_{p^*}$, $\Omega \in V^*$ with $D := L^* \Omega \neq 0$, and suppose that for a sequence $s_n \rightarrow 0^+$,*

$$q^*(a + s_n D) \leq s(2s_n) \quad \text{and} \quad N_m(w_m - s_n \Omega_m) \leq s(2s_n) \quad (m \in I).$$

Then $\sum_{j \notin F} (|D_j| - z_j D_j) = 0$: off F , D is supported in the contact set K with $z_j D_j = |D_j|$. Since $D \in Y \setminus \{0\}$ is not finitely supported, $F \cup K$ is infinite. In particular there are no such transfers when $a \in c_{00}$ and K is finite (e.g. at every norm-attaining f).

Proof. By Lemma 3.3 (dropping nonnegative terms), $q^*(a + sD) \geq 1 + sD(\hat{z}) + s \sum_{j \notin F} (|D_j| - z_j D_j)$ for $s > 0$, and $N_m(w_m - s\Omega_m) \geq \langle w_m - s\Omega_m, \zeta_m \rangle / \sigma_m = 1 -$

$s\langle\Omega_m, \zeta_m\rangle/\sigma_m$. With $s = s_n$, multiply the first inequality by q_0 and the m -th by σ_m and add: since $q_0 D(\hat{z}) = D(\xi) = \langle\Omega, L^{**}\xi\rangle = \sum_m \langle\Omega_m, \zeta_m\rangle$ and $q_0 + \sum_m \sigma_m = 1$,

$$q_0 s_n \sum_{j \notin F} (|D_j| - z_j D_j) \leq s(2s_n) - 1 \leq 2s_n^2.$$

Divide by s_n and let $n \rightarrow \infty$. Each term $|D_j| - z_j D_j$ is nonnegative and vanishes only if $D_j = 0$ or $|z_j| = 1$ with $z_j D_j = |D_j|$. $\square$

Corollary 6.10 (Two-sided linear decompositions). *Let $g \in \mathcal{C}(f)$ and suppose there are $\tau > 0$ and $(B_\pm, \Omega_\pm) \in \ell_1 \times V^*$ with $g = B_\pm + L^* \Omega_\pm$, $\max(q^*(a + tB_+), \|w + t\Omega_+\|) \leq s(t)$ for $0 \leq t \leq \tau$ and $\max(q^*(a + tB_-), \|w + t\Omega_-\|) \leq s(t)$ for $-\tau \leq t \leq 0$. If $F \cup K$ is finite, then $B_+ = B_-$, $\Omega_+ = \Omega_-$, and $g \in \text{cl Cert}(f)$.*

Proof. For $0 < t \leq \tau$ average the decompositions of $f + tg$ and $f - tg$: $f = (a + \frac{t}{2}D) + L^*(w - \frac{t}{2}\Omega_D)$ with $D := B_+ - B_-$, $\Omega_D := \Omega_- - \Omega_+$, $L^*\Omega_D = D$, and both component norms are at most $s(t)$ by convexity. If $D \neq 0$, Proposition 6.9 with $s_n = t_n/2$ contradicts the finiteness of $F \cup K$. Hence $D = 0$, and $\Omega_+ = \Omega_-$ by injectivity of L^* . The common decomposition makes g locally split, and Theorem 4.10 applies. $\square$

Remark 6.11 (Resonance types; the sign rule). The transfer in the proof of Corollary 6.10 exists for every mate and every pair of admissible decompositions at scales $\pm t$; on the contacts, for instance, the cheaply used parts of B_+ and of B_- have opposite signs, so they add up in $D = B_+ - B_-$. (A general “matching at scale” statement for all one-sided resources is only available as a sketch and is not used.) A mate outside $\mathcal{K}(f)$ therefore needs *resonances*: cheap transfers with large one-sided mass at arbitrarily small scales. Proposition 6.9 shows that exact (scale-free) resonances require infinitely many contacts (or near-flip coordinates of F when $a \notin c_{00}$); Theorem 6.6 realizes one. Approximate resonances run through one-sided block carriers: weak peaks and near-threshold strict non-peaks at depth comparable to the scale. For these we only record an elementary fact: if $u_{k,m}(\xi) \neq 0$, then $\text{sign } w_m(k) = \text{sign } u_{k,m}(\xi)$ (on P_m by Lemma 1.6, off P_m because $w_m(k) = C_m \zeta_m(k)/(\Phi_m(k)^2 \sigma_m)$). Consequently a one-sided usage of (k, m) pushes in the direction $-\text{sign } u_{k,m}(\xi)$, and two near-threshold carriers whose vectors differ by $o(\Phi)$ in the ξ -direction (for instance Martín’s cross-block near duplicates with small perturbations) serve the same side of $t = 0$ and cannot switch with each other. This is a heuristic guide; no classification of approximate resonances is proved here. When the gaps of the carriers are not summable, Theorem 4.12 puts the corresponding mates in $\text{cl Cert}(f)$; the summable case and the weak-peak case are open (Remark 4.13).

7. ENGINEERED APPROXIMANTS

Proposition 1.11 shows that the coordinates of the base contact of a norm-attaining approximant outside a finite window are free. This section will treat recovery of resonant mates, such as the switching mates of Theorem 6.6, along norm-attaining approximants that are engineered with this freedom, rather than along the canonical truncations.

8. STATUS OF THE DENSITY PROBLEM

8.1. What is proved. Let T be an arbitrary admissible operator. Density of $\text{NA}((c_0, p), \ell_2^2)$ is equivalent to $\mathcal{R} = S_{p^*}$ (Proposition 1.12), and by [7, Remark

martin-tail] it suffices to prove $\mathcal{R} = S_{p_N}^*$ for arbitrarily large N . A first row f belongs to $\mathcal{R}$ under each of the following sufficient conditions:

- (S1) f attains its norm (Proposition 1.12(d));
- (S2) f belongs to the residual set Ω of Hausdorff continuity points of the fibres [6, Theorem 2.2];
- (S3) $\mathcal{C}(f) = \text{cl Cert}^{\text{sh}}(f)$; then every mate is recovered along every sequence $f_n \rightarrow f$ (Theorem 3.13); more generally $\mathcal{C}(f) = \mathcal{K}(f)$ when $a \in c_{00}$ (Proposition 6.1);
- (S4) I is finite and the normer of f is C-tame without degenerate peaks (Theorem 5.13);
- (S5) f is a limit of points of $\mathcal{R}$ along which the support functions $\tilde{r}$ are asymptotically at least those of f (Theorem 2.4).

For an individual mate $g \in \mathcal{C}(f)$, $(f, g) \in \text{cl NA}$ whenever g lies in $\text{cl Cert}^{\text{sh}}(f)$; this covers finite and shifted certificates (Theorems 3.8, 3.13), weighted directions with $O(\sigma)$ box tails (Theorem 4.6), locally split mates and split-contractive operators (Theorem 4.10, Corollary 4.11), every mate that is a certificate of radius $\gtrsim t$ up to an $O(t)$ remainder at all small scales (Corollary 4.2), and carrier mates with non-summable gaps (Theorem 4.12). When $a \in c_{00}$, also every balanced finite certificate with $\Gamma_w \leq 1$ is recovered (Theorem 5.8). The analogous statements for ℓ_2^{d+1} hold for joint certificates (Theorem 3.15).

On the negative side, the intrinsic programme cannot succeed in general: for some admissible T there is a non-attaining f at which the defect $\mathcal{C}(f) \setminus \mathcal{K}(f)$ is infinite dimensional (Theorem 6.6), and its elements are not recovered along the canonical truncations (Proposition 6.8). At every point where $a \in c_{00}$ and K is finite (in particular at every norm-attaining point), exact resonances do not exist (Proposition 6.9). None of this is a counterexample to density: a counterexample would produce a two-dimensional Hilbert space without Lindenstrauss' property B, answering Question 6 of Johnson and Wolfe [2] negatively.

8.2. What remains. By Lemma 1.15 and Corollary 3.9(c), the density problem for p_N is equivalent to the following statement, in which only the defect matters.

Problem 8.1. Let $N \geq 1$, let T be admissible, and let $f \in S_{p_N}^*$. Is it true that for every $g \in \text{Def}(f)$ and every $\rho < 1$ there are norm-attaining $f_n \rightarrow f$ with $\rho g \in \text{Li}_n \mathcal{C}(f_n)$? Equivalently, is every pair $(f, \rho g)$ with $g \in \mathcal{C}(f) \setminus \text{cl Cert}^{\text{sh}}(f)$ a limit of norm-attaining operators into ℓ_2^2 ?

A positive answer for arbitrarily large N gives density for Martín's norm (Remark 1.4); a negative answer for one N and one admissible T gives a two-dimensional range without property B. By Proposition 6.8, the approximants in Problem 8.1 must in general depend on the mate and cannot be the canonical truncations. The following more specific questions are open.

Problem 8.2 (Resonant mates). Recover the switching mates of Theorem 6.6 along engineered norm-attaining approximants; more generally, prove a recovery theorem for mates whose admissible decompositions are, on each side of $t = 0$, a certificate plus an exact-resonance part plus $O(t)$. The case of two-piece mates whose block carriers have different first-order coefficients on the two sides ($d_- < d_+$) is not covered by any known argument.

Problem 8.3 (Second-order defect). Is there f with $a \in c_{00}$ and a balanced finite certificate $g \in \mathcal{C}(f) \cap S(f)$ with $\Gamma_w(g) > 1$? By Proposition 5.12 such an f must have degenerate peaks, infinitely many strict non-peaks, failure of (MS), or infinitely many contacts; in the last case a finite-dimensional analogue suggests that the second-order coefficient can drop below Γ_w (Remark 5.14).

Problem 8.4 (Approximate resonances). Do switching mates carried, on both sides of $t = 0$, by near-threshold strict non-peaks with summable gaps or by weak peaks belong to $\text{cl Cert}(f)$? There is no linear obstruction for them (Remark 4.13); a proof of non-membership would require a sequence $f_n \rightarrow f$ with $g \notin \text{Li}_n \mathcal{C}(f_n)$.

Problem 8.5 (Universality of the defect). Does *every* admissible T (in particular every T produced by the proof of Martín’s Lemma B) admit a first row with nonempty defect? The linear obstruction requires a first row with few strict non-peaks, which an adversarial T may prevent.

Finally we note a gap between necessary and sufficient conditions in the reduction itself: density into ℓ_2^{d+1} implies that tuples in $\mathcal{C}_d(f)/\sqrt{d}$ are recovered along a common sequence, but we do not know whether density into ℓ_2^2 alone forces the existence, for each f , of one norm-attaining sequence along which the whole fibre $\mathcal{C}(f)$ is recovered (condition (a) of Corollary 2.3); only the sufficiency of that condition is used.

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